# *Rotational Dynamics*

## Rotational analog for:

→ inertia

→ Newton's 2nd law

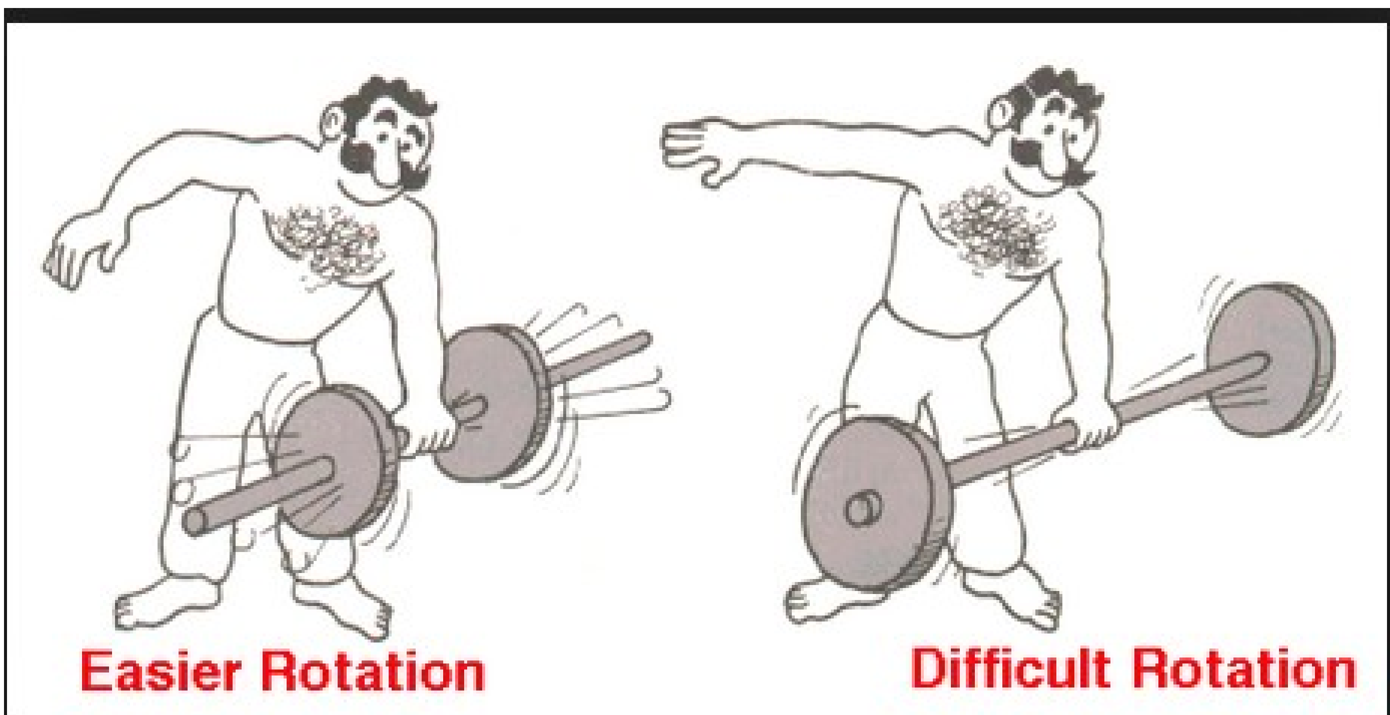
→ Kinetic energy

→ Angular momentum

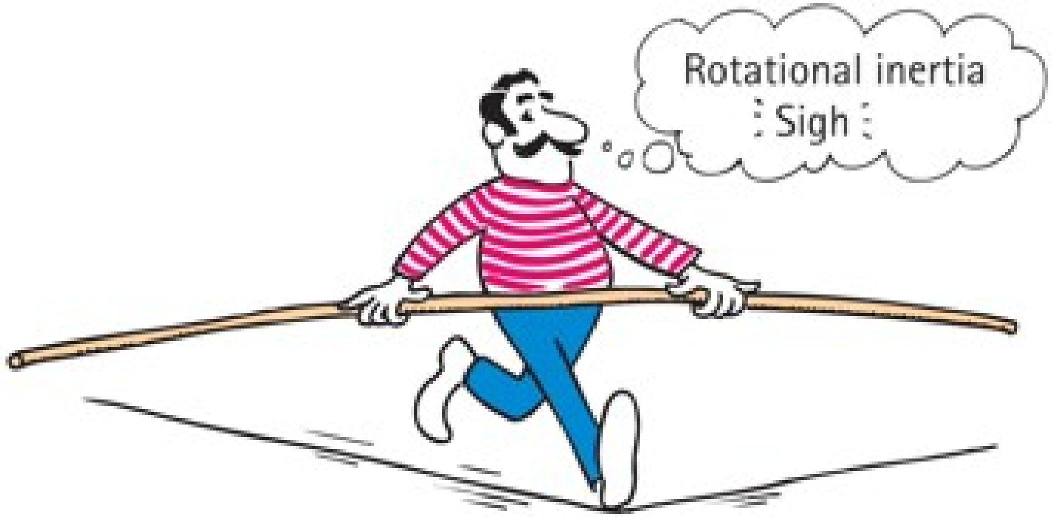
http://hyperphysics.phy-astr.gsu.edu/hbase/circ.html

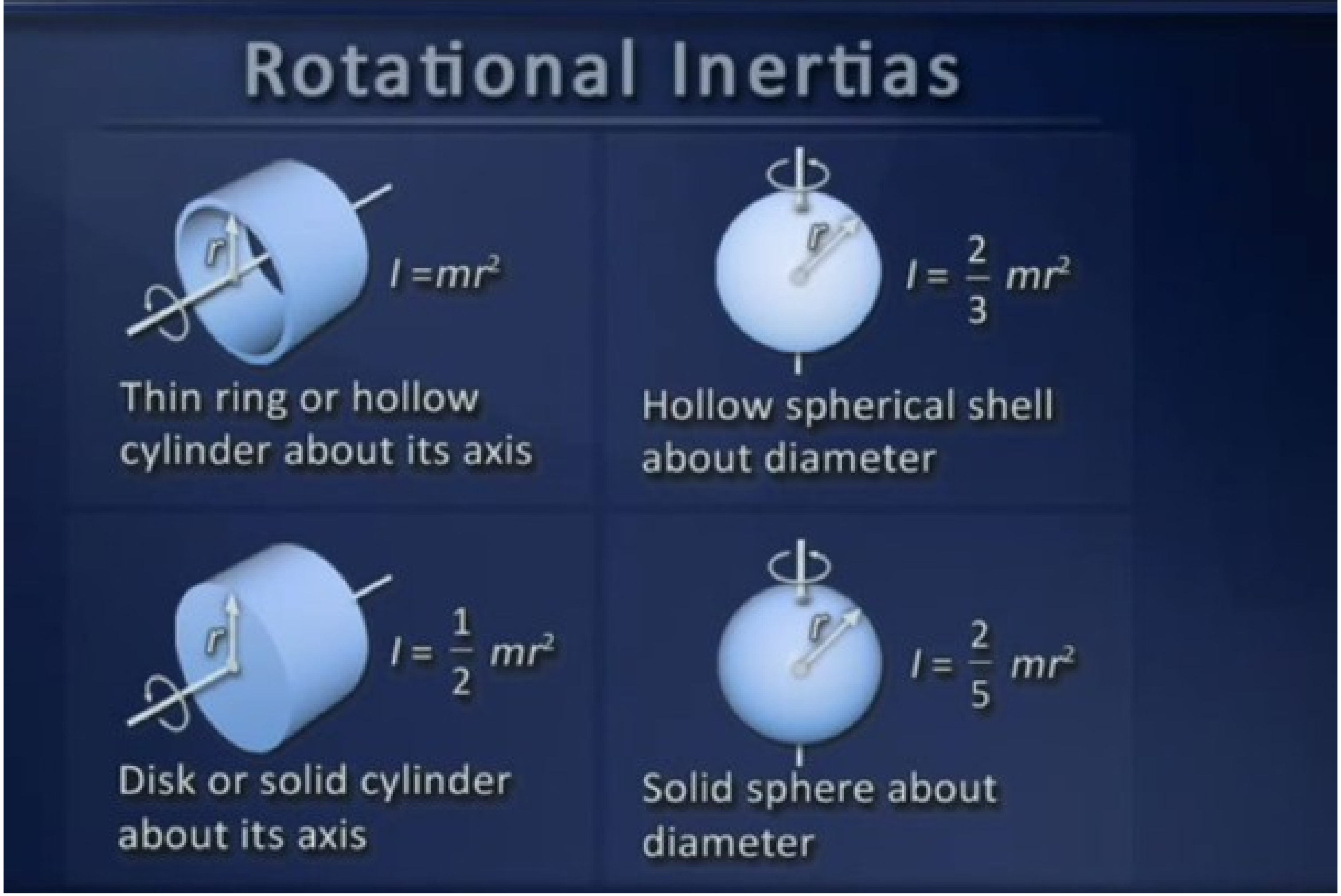
# Rotational analog for mass moment of inertia

**Rotational inertia ( *I* ):** resistance to changes in rotational motion



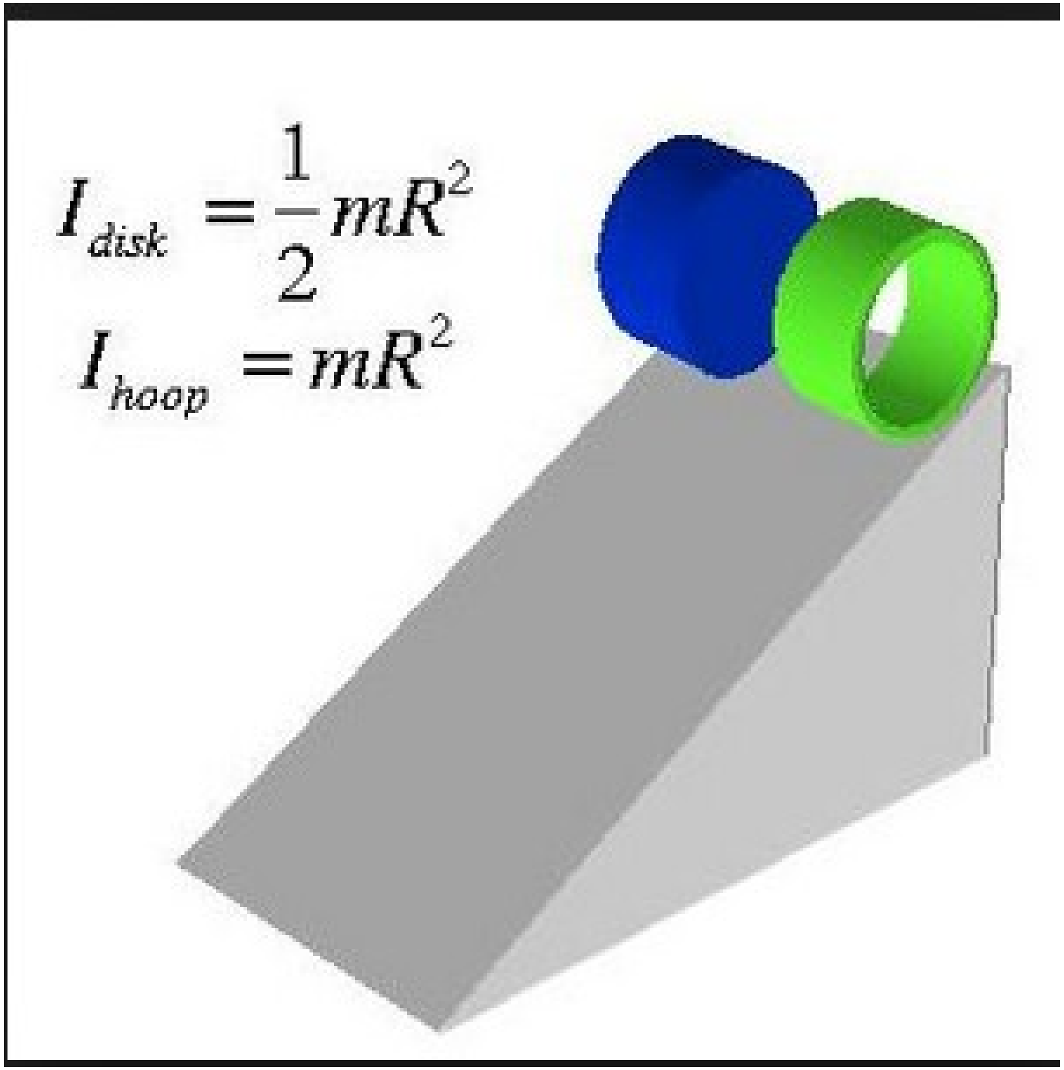
- The greater the rotational inertia, the harder it is to change its rotational state.
  - A tightrope walker carries a long pole that has a high rotational inertia, so it does not easily rotate.
  - Keeps the tightrope walker stable.






Moment of inertia/rotational depends on the DISTRIBUTION of mass around the axis of rotation.

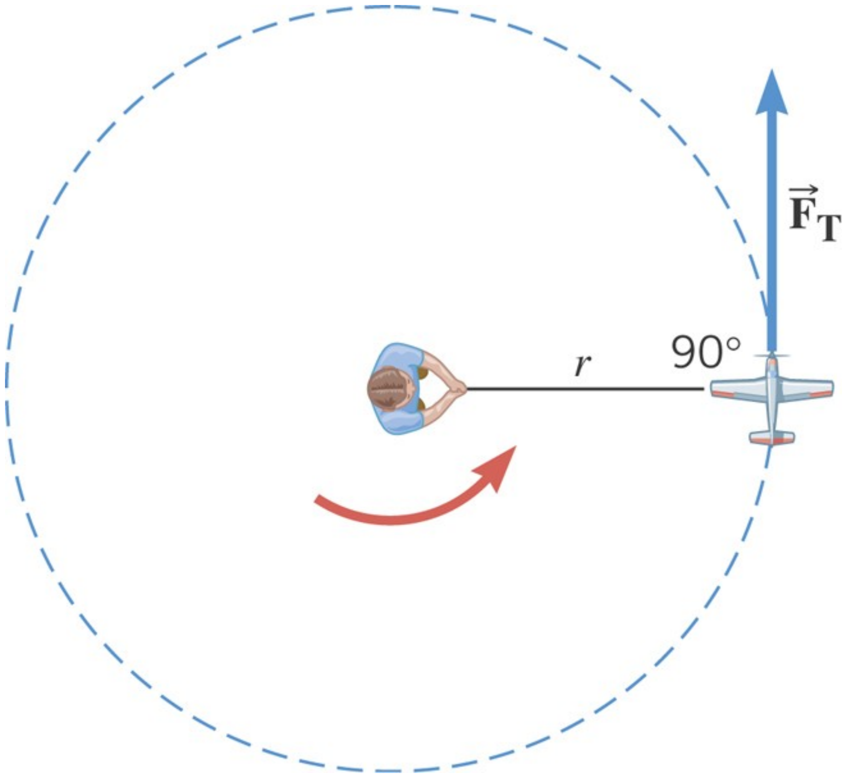
For solid objects about axis → inertia = coeff x mass $r^2$




Movie race of the objects.
Its “falling “ so mass does not matter. Interestingly, R does not matter.
We can find the acceleration using the conservation of energy.

https://www.youtube.com/watch?v=beCUQftpSqE

***Rotational analog for mass = moment of inertia***



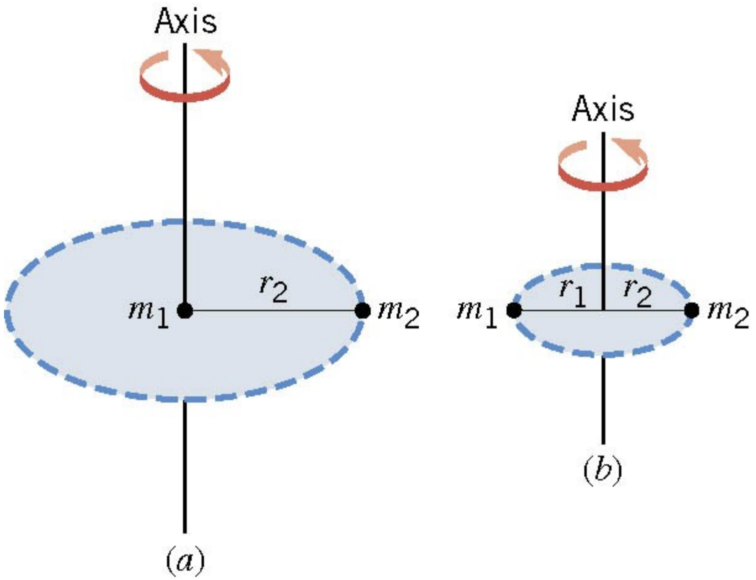

$$I = m\,r^2$$

For a point mass

Moment of Inertia, I

## *Example 9* The Moment of Inertial Depends on Where the Axis Is.

Two particles each have mass and are fixed at the ends of a thin rigid rod. The length of the rod is *L*. Find the moment of inertia when this object rotates relative to an axis that is perpendicular to the rod at
(a) one end and (b) the center.

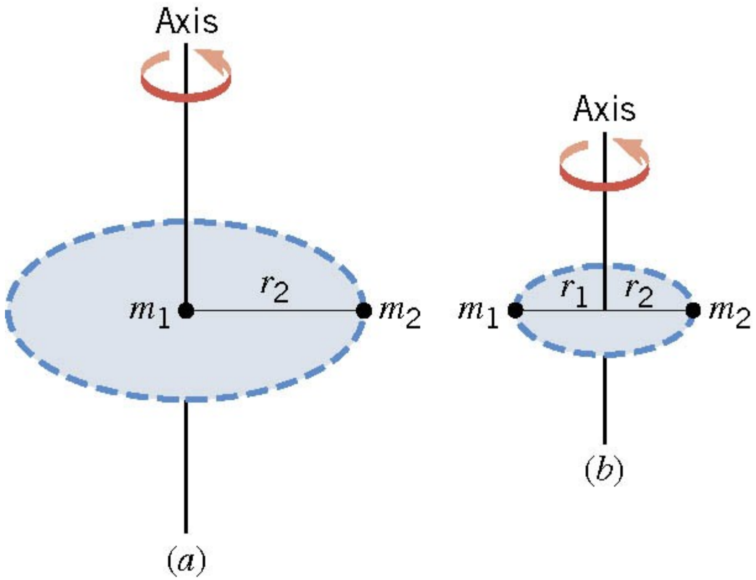


# 9.4 Newton's Second Law for Rotational Motion About a Fixed Axis

(a) $$I = \sum \left(mr^2\right) = m_1 r_1^2 + m_2 r_2^2 = m\left(0\right)^2 + m\left(L\right)^2$$

$$m_1 = m_2 = m \qquad r_1 = 0 \quad r_2 = L$$

$$I = mL^2$$

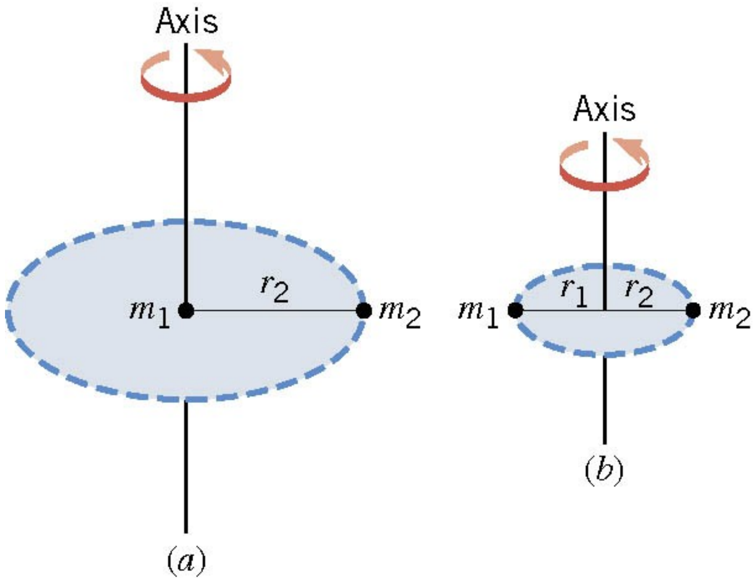


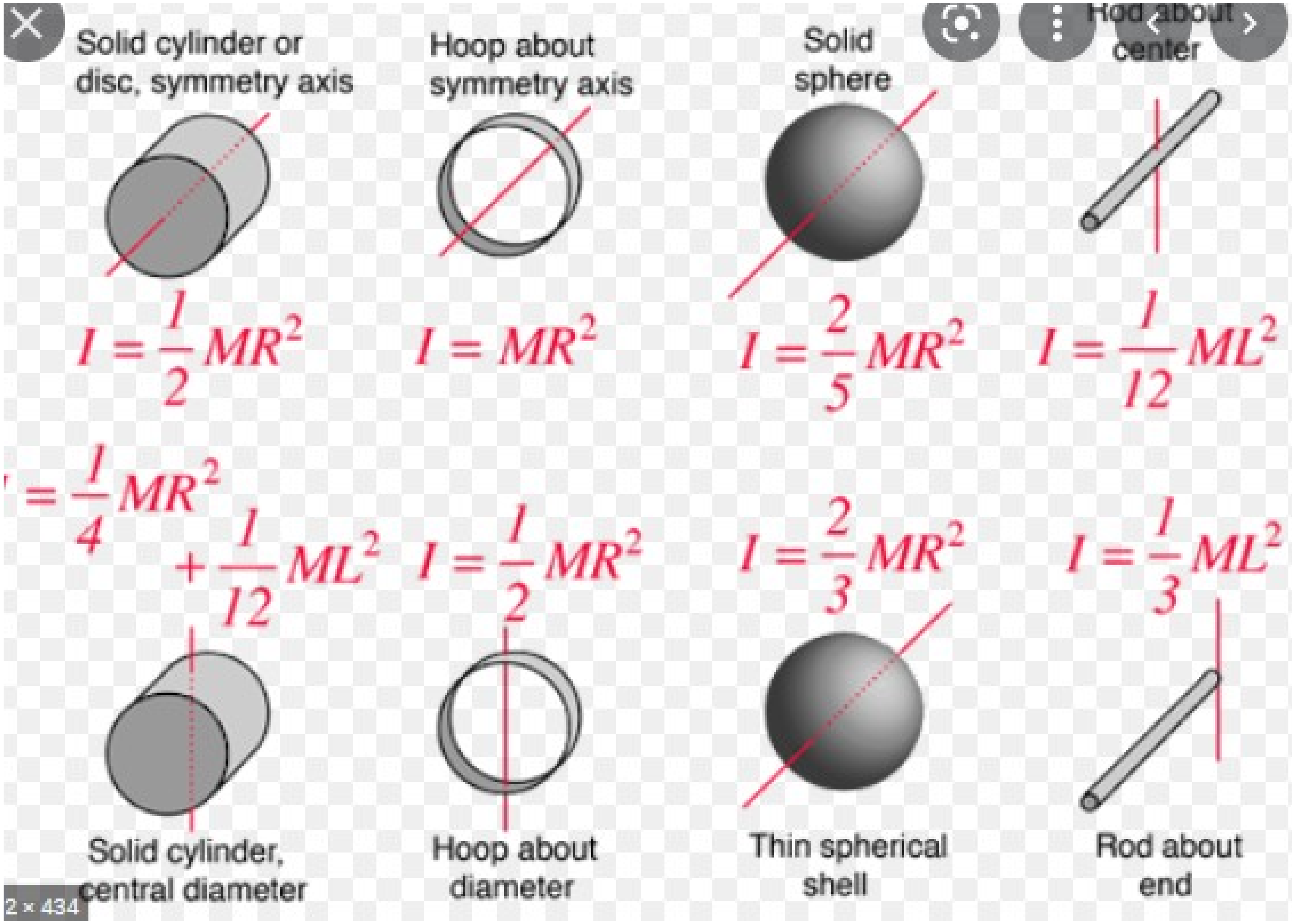
# 9.4 Newton's Second Law for Rotational Motion About a Fixed Axis

(b) $I = \sum \left(mr^2\right) = m_1 r_1^2 + m_2 r_2^2 = m\left(L/2\right)^2 + m\left(L/2\right)^2$

$m_1 = m_2 = m$ $\qquad r_1 = L/2 \quad r_2 = L/2$

$I = \frac{1}{2} m L^2$






http://hyperphysics.phy-astr.gsu.edu/hbase/mi.html

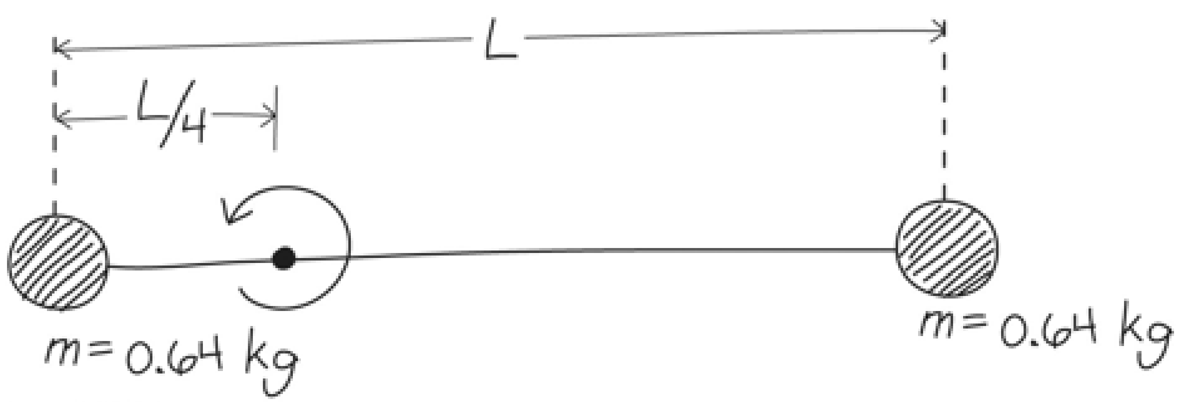
# Clicker Question 2

- Consider the dumbbell in the figure. How would its rotational inertia change if the rotation axis were at the center of the rod?

A. *I* would increase.

B. *I* would decrease.

C. *I* would remain the same.



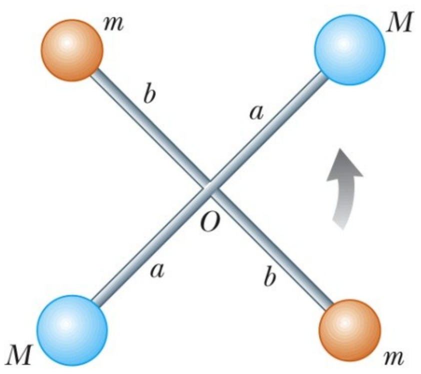
## Clicker Question 8

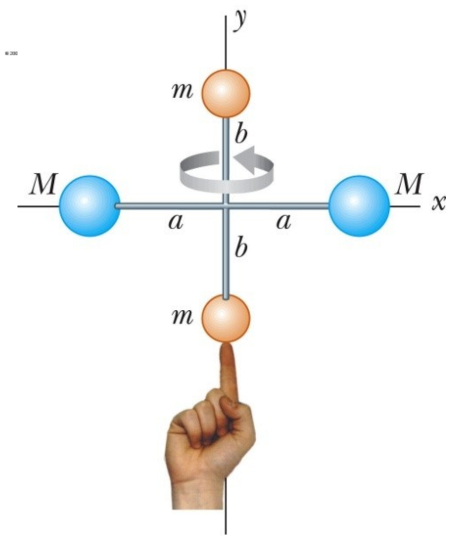
- Two uniform solid spheres have the same radius, but one is three times more massive than the other. What is the ratio of the larger moment of inertia to that of the smaller moment of inertia?

A. 3
B. 6
C. 9
D. 16
E. 27

# The Baton Twirler

- Consider an unusual baton made up of four sphere fastened to the ends of very light rods. Each rod is 1.0m long ($a = b = 1.0$ m). $M$ = 0.3 kg and $m$ = 0.2 kg.
- (a) Find $I$ about an axis perpendicular to the page and passing through the point where the rods cross.
- (b) The majorette tries spinning her strange baton about the axis $y$, calculate $I$ of the baton about this axis



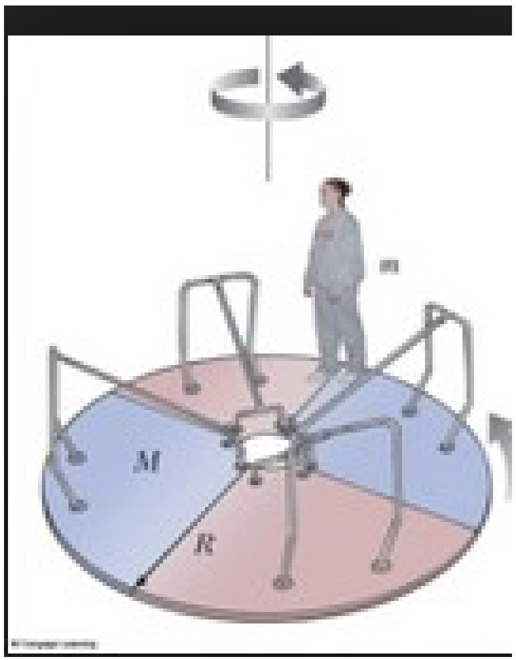



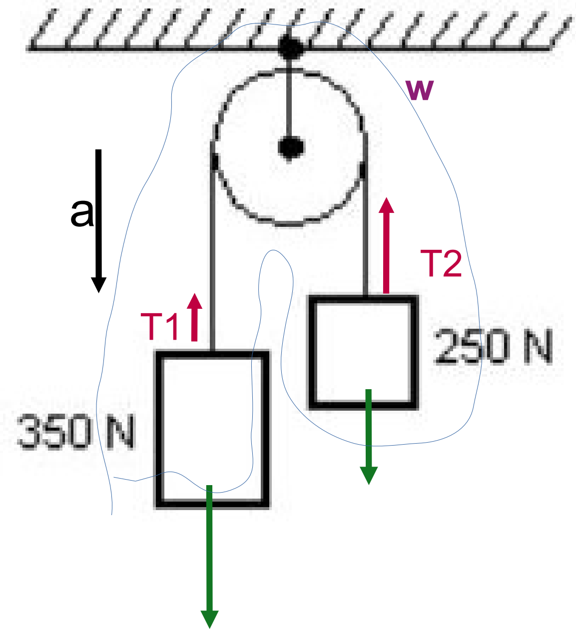
(a)



Textbook 9.6

The moment of inertia of a rigid composite system is the sum of the moments of inertia of its component subsystems (all taken about the same axis).

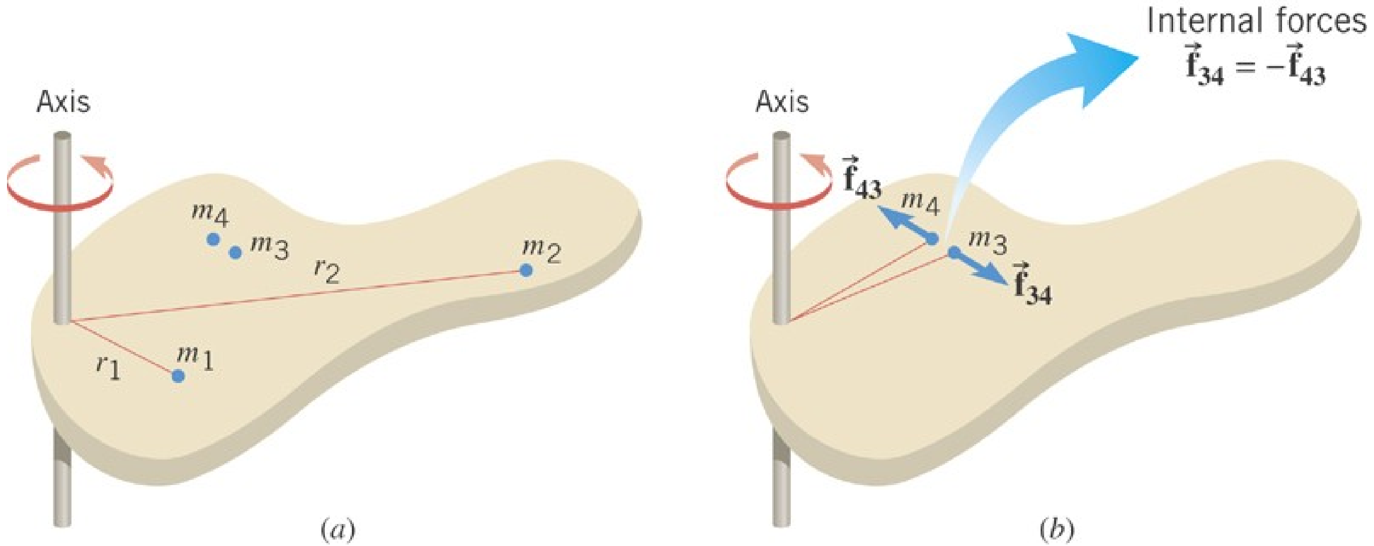



The mass of the person is 60Kg
The pass the of turning platform is 500Kg
The radius is 2 m
What is the total inertia?

Consider the system =
Pulley of mass M, radius R (plain cylinder) and the 2 masses 25kg and 35 kg.
Its like the masses are glue to The pulley (forget about the string). What is the total inertia?

*Finding the moment of inertia using calculus*




$$I = \sum \left( m r^2 \right)$$

Moment of inertia

$$I1 = \left( m_1 r_1^2 \right)$$

$$I2 = \left( m_2 r_2^2 \right)$$

$$\vdots$$

$$I = \left( m_N r_N^2 \right)$$

# Moment of Inertia of Extended Objects

- Divided the extended objects into many small volume elements, each of mass $\Delta m_i$
- We can rewrite the expression for $I$ in terms of $\Delta m$

$$I = \lim_{\Delta m_i \to 0} \sum_i r_i^2 \Delta m_i = \int r^2 dm$$

- Consider a small volume such that $dm = \rho\, dV$. Then

$$I = \int \rho r^2 dV$$

- If $\rho$ is constant, the integral can be evaluated with known geometry, otherwise its variation with position must be known

# Densities

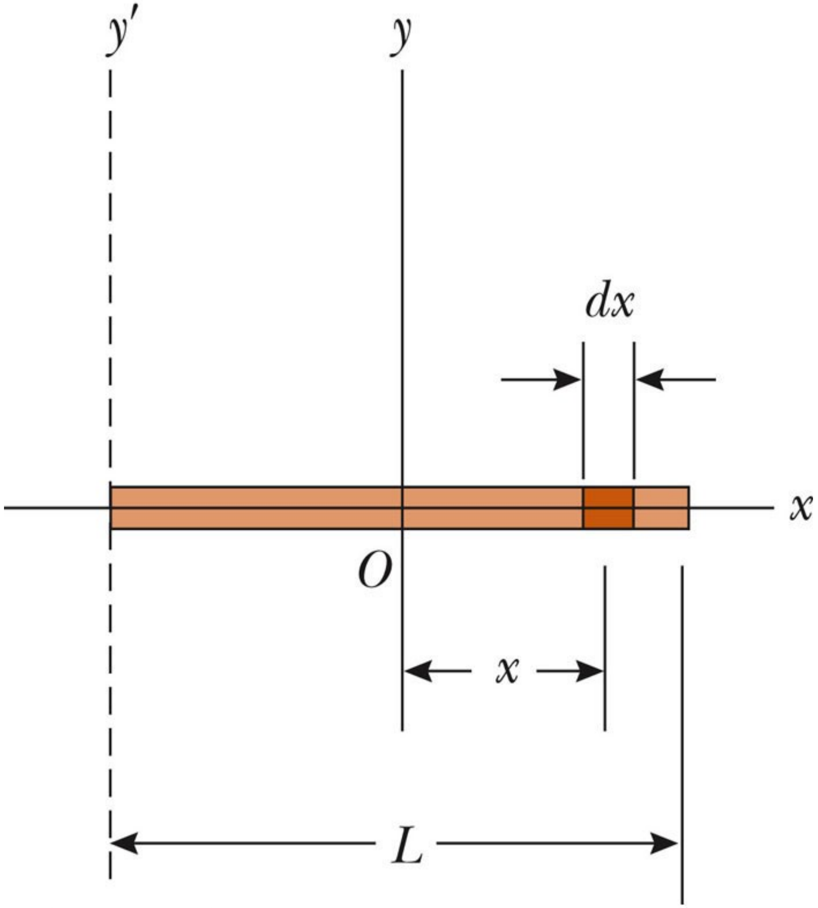
- You know the density (volume density) as mass/unit volume
  - $\rho = M/V = dm/dV \qquad => dm = \rho dV$
- We can define other densities such as surface density (mass/unit area)
  - $\sigma = M/A = dm/dA \qquad => dm = \sigma dA$
- Or linear density (mass/unit length)
  - $\lambda = M/L = dm/dx \qquad => dm = \lambda dx$

# Moment of Inertia of a Uniform Rigid Rod

- The shaded area has a mass
  - $dm = \lambda\, dx$
- Then the moment of inertia is

$$I_y = \int r^2 dm = \int_{-L/2}^{L/2} x^2 \frac{M}{L}\, dx$$

$$I = \frac{1}{12} ML^2$$

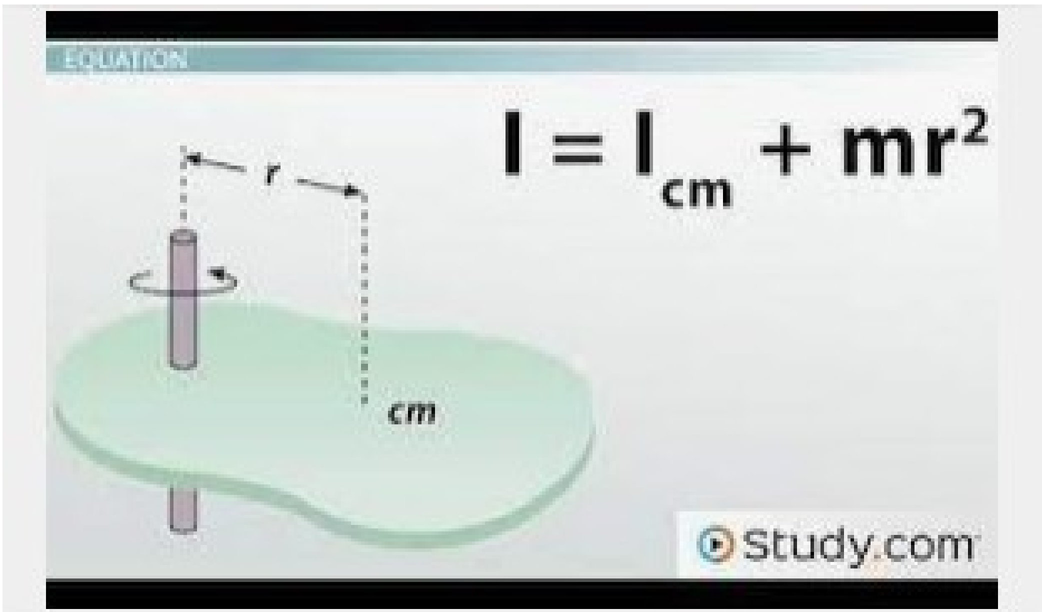





**Textbook 9.10, 9.11 → important for engineers**

How to compute I relative to the axis y’? We know I relative to y
That goes through the CM (refer to previous slide)

Use the parallel axis theorem




The Parallel-Axis Theorem & the Moment ...

Textbook 9.9

http://hyperphysics.phy-astr.gsu.edu/hbase/parax.html

# Rotational dynamics.
# There is a symmetry with linear dynamics.

Newton's $2^{nd}$ law: $F = ma$

Rotational analog of Newton's $2^{nd}$ law: $\tau = I\alpha$ For solid around axis

Torque = r x F (cross product)

Kinetic energy: $K = \frac{1}{2}mv^2$

Rotational kinetic energy: $K_{rot} = \frac{1}{2}I\omega^2$

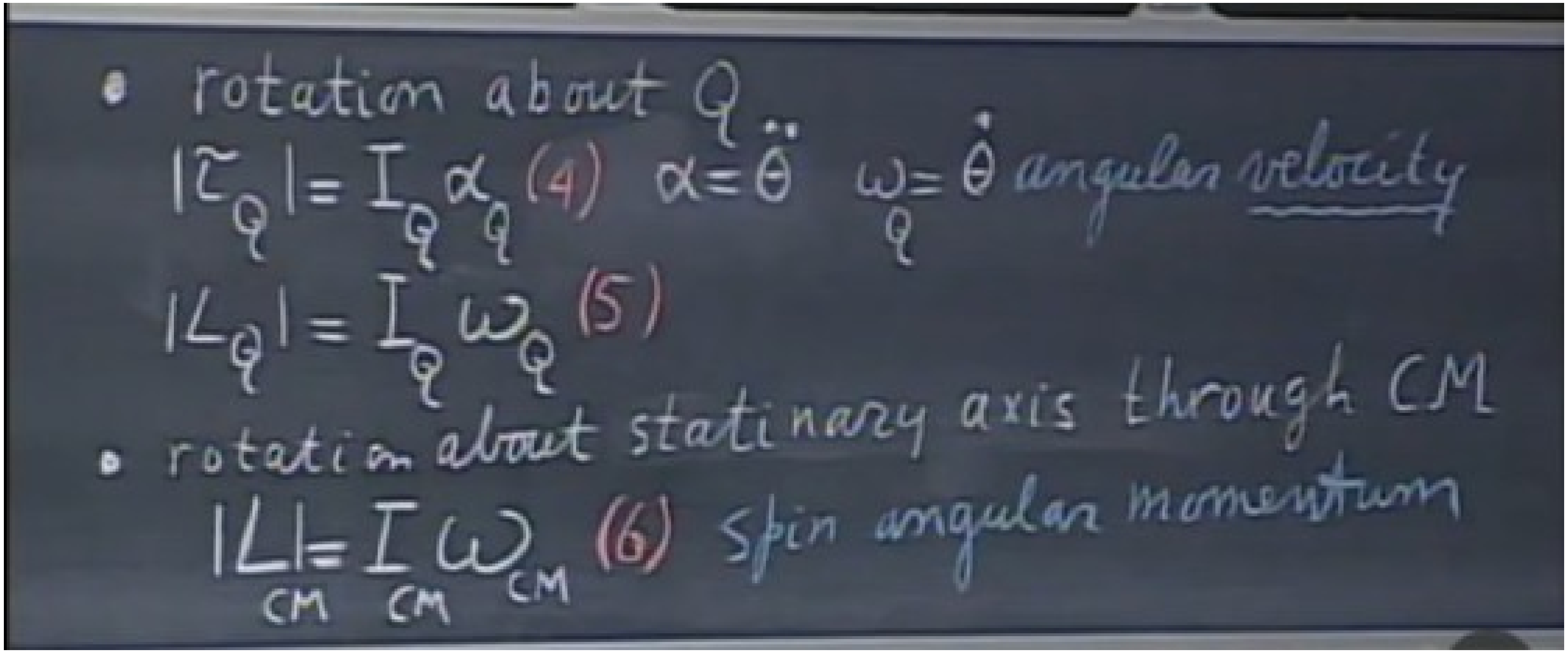
Momentum: $p = mv$

For solid around axis

Angular momentum: $L = I\omega$

L = r x p (cross product)
For a point mass

**ALSO F = dp/dt and torque = dL/dt**




Excellent lecture online (Walter Lewin).

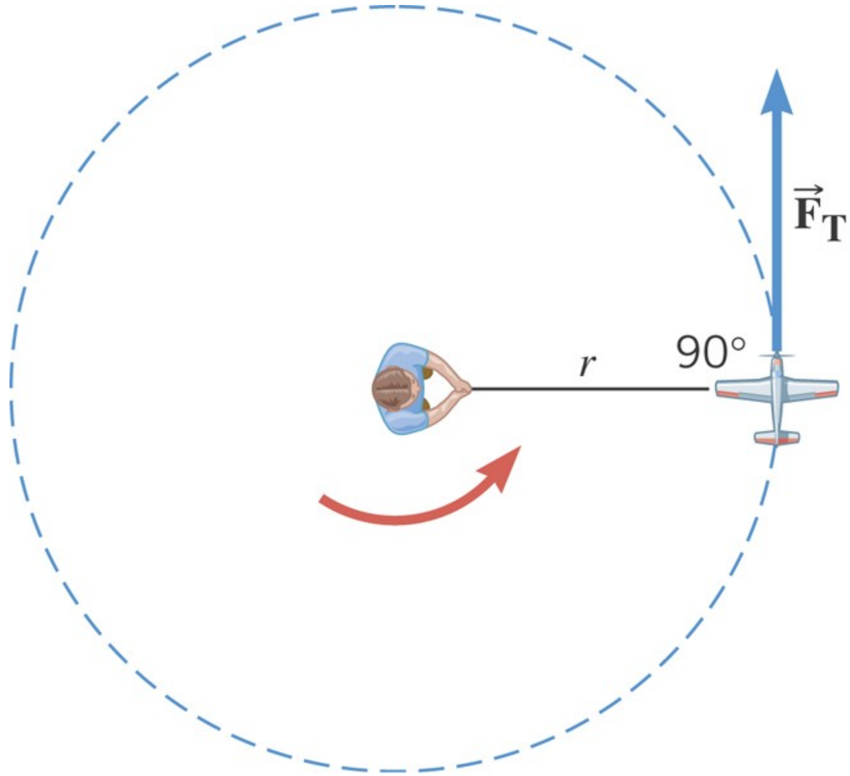
https://www.youtube.com/watch?v=4Mw6oD1MnKg

## *Analog for Newton's 2nd law*

$$F_T = m a_T$$

$$a_T = r\alpha$$

$$\tau = F_T r$$




$$\tau = \left(m r^2\right) \alpha$$

Moment of Inertia, I

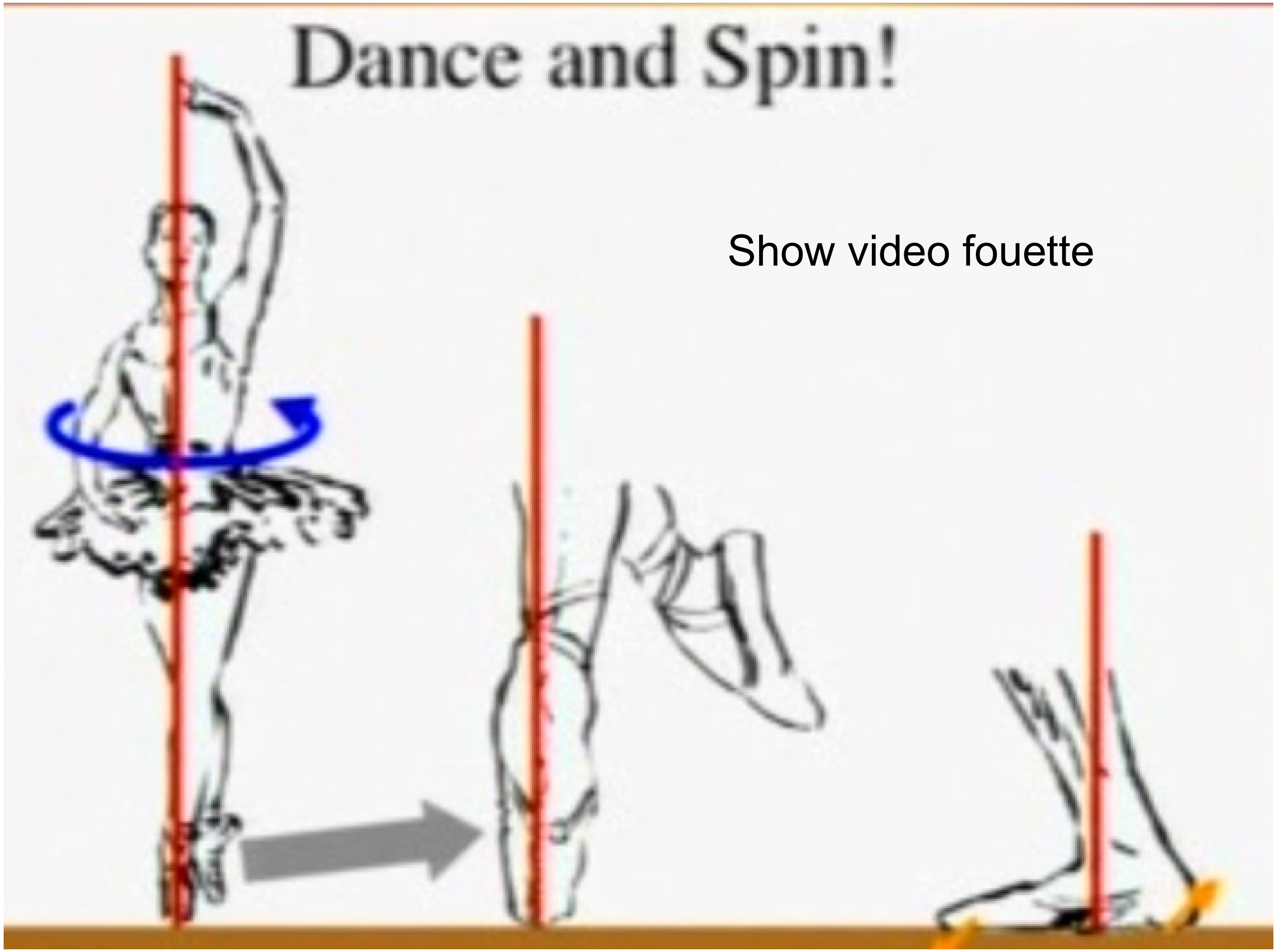
## ROTATIONAL ANALOG OF NEWTON'S SECOND LAW FOR A RIGID BODY ROTATING ABOUT A FIXED AXIS

Net external torque/axis = (moment of inertia/axis) x (rotational acceleration)

$$\sum \tau = I \alpha$$

$$I = \sum \left( mr^2 \right)$$

***Requirement:*** Angular acceleration must be expressed in radians/s$^2$.




Show video fouette

https://www.youtube.com/watch?v=EixtGKNP6ok
028

http://www.physics.unlv.edu/~jeffery/astro/moon/tidal_locking_origin.html

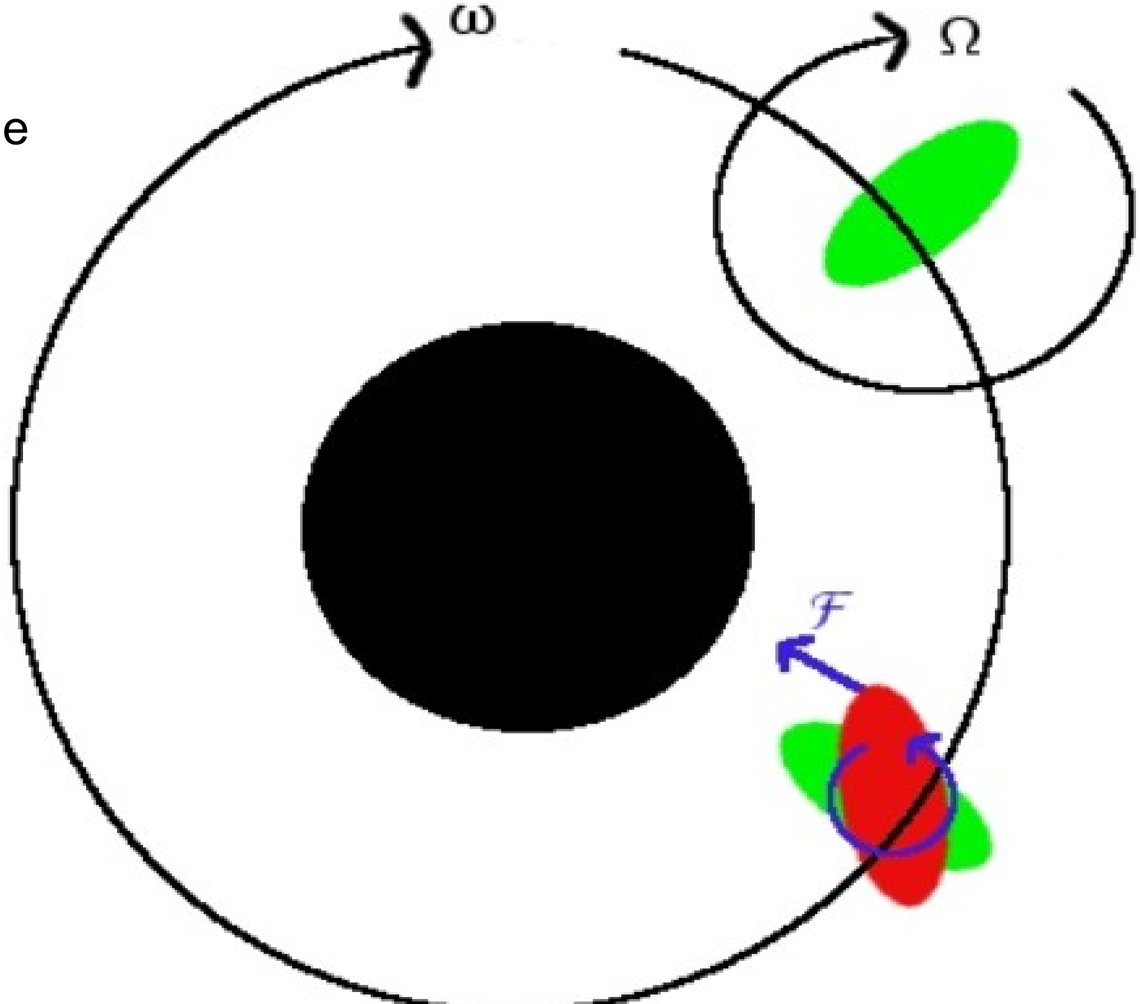
The angular speed are equal

The torque from the Earth on the Moon

Locks the Moon.

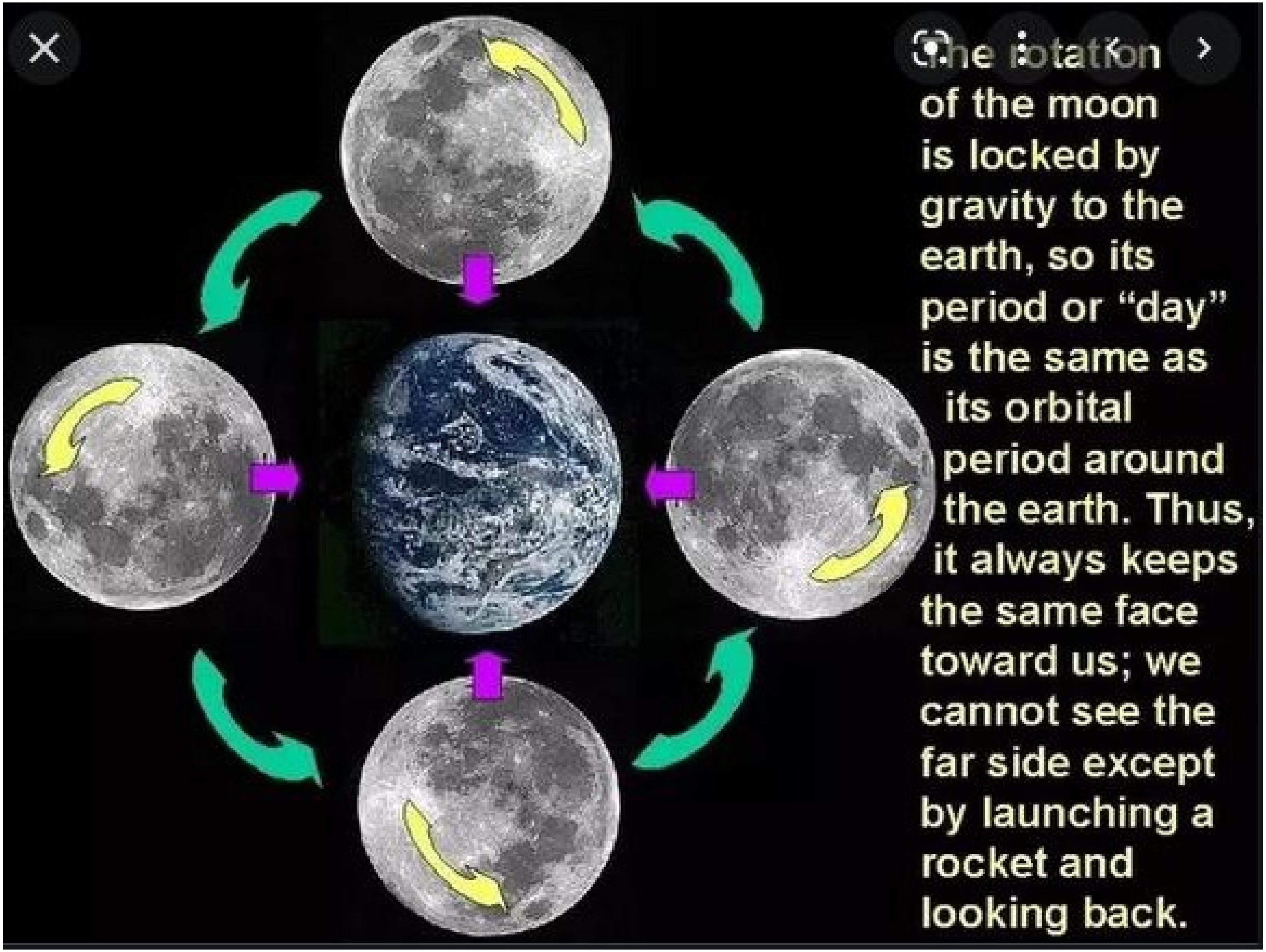
We always sees the same

Face.

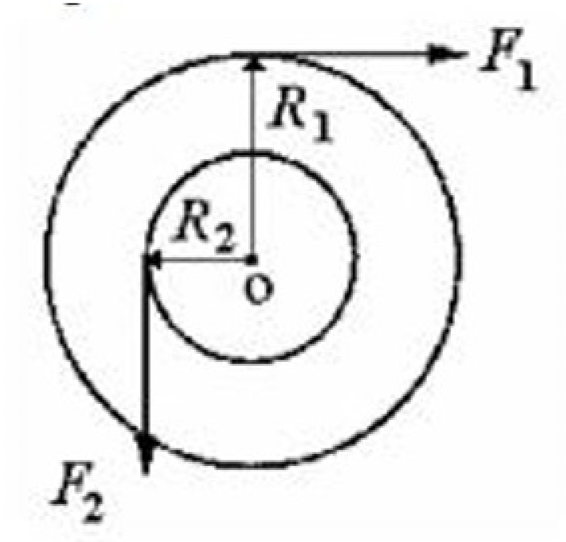



The Earth and the Moon
are locked together

Same period for
Rotation around the Earth
Or for the rotation of
The Moon about its axis.


of the moon is locked by gravity to the earth, so its period or "day" is the same as its orbital period around the earth. Thus, it always keeps the same face toward us; we cannot see the far side except by launching a rocket and looking back.

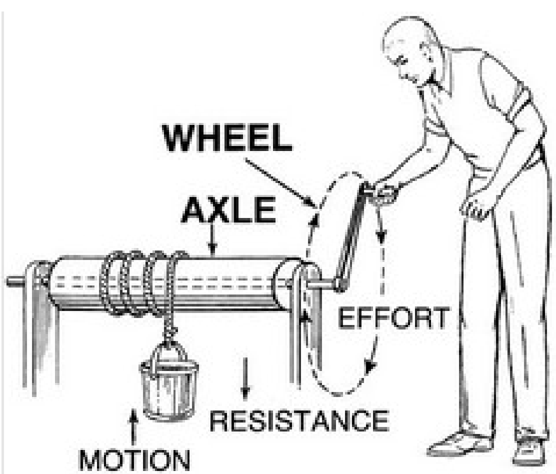
# Example: wheel and axle → its a machine



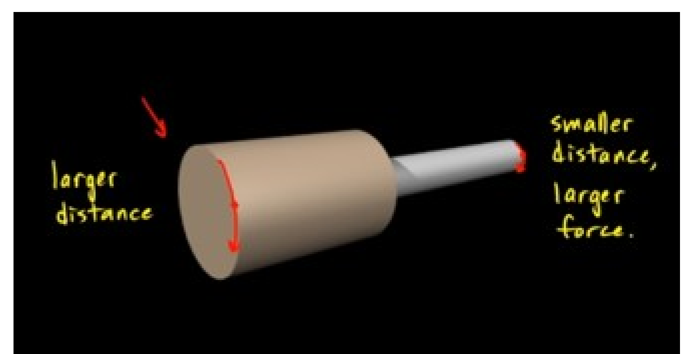

**Write Newton's second law**
**Sum(torque) = Inertia x acceleration**

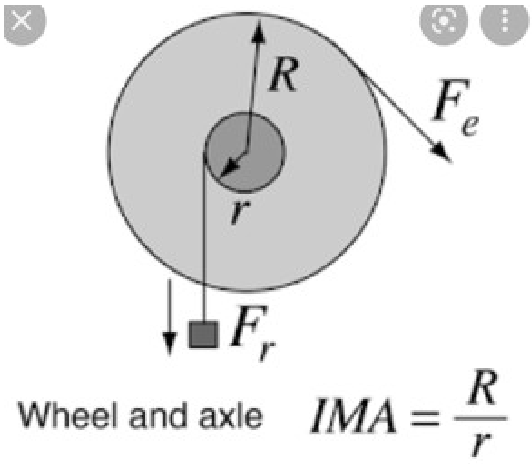
**F1 R1 – F2  R2 = I x acceleration**
**I is the inertia of the pulley (cylinder 0.5 $MR^2$)**

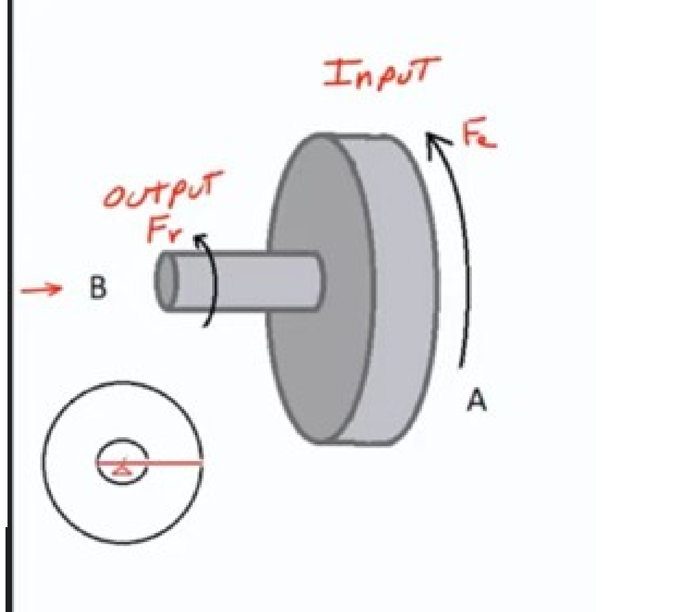
If equilibrium → F1 R1 – F2  R2 = 0
Say F1 is the effort and F2 is the load and we are pulling at a constant speed.
What is the mechanical advantage ? (F1/F2 ?)

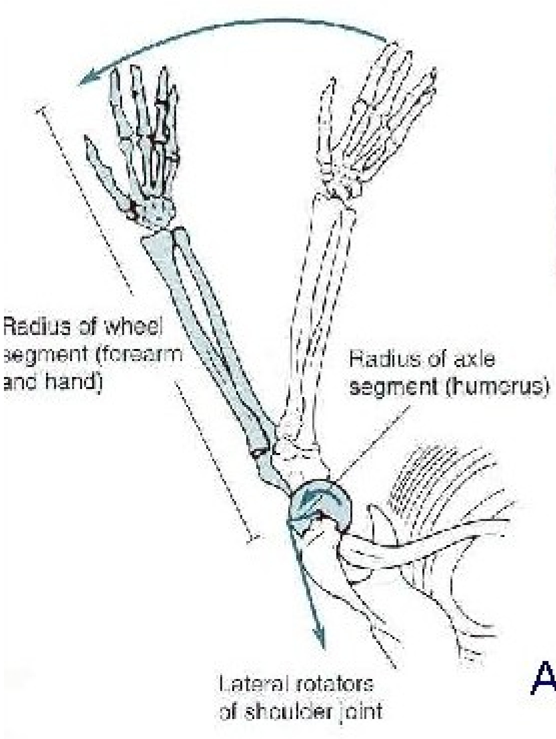


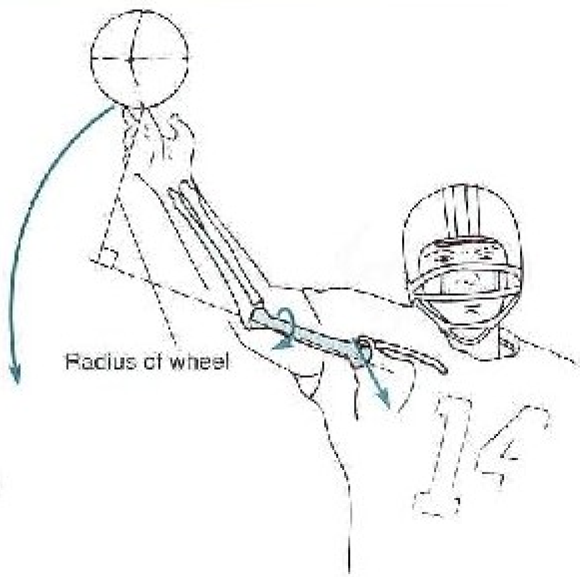

Big radius x small force = small radius x big force
That means: (same angle swept)
Big distance x small  effort = small distance x large load
(distance = radius x angle swept)
→ conservation of energy





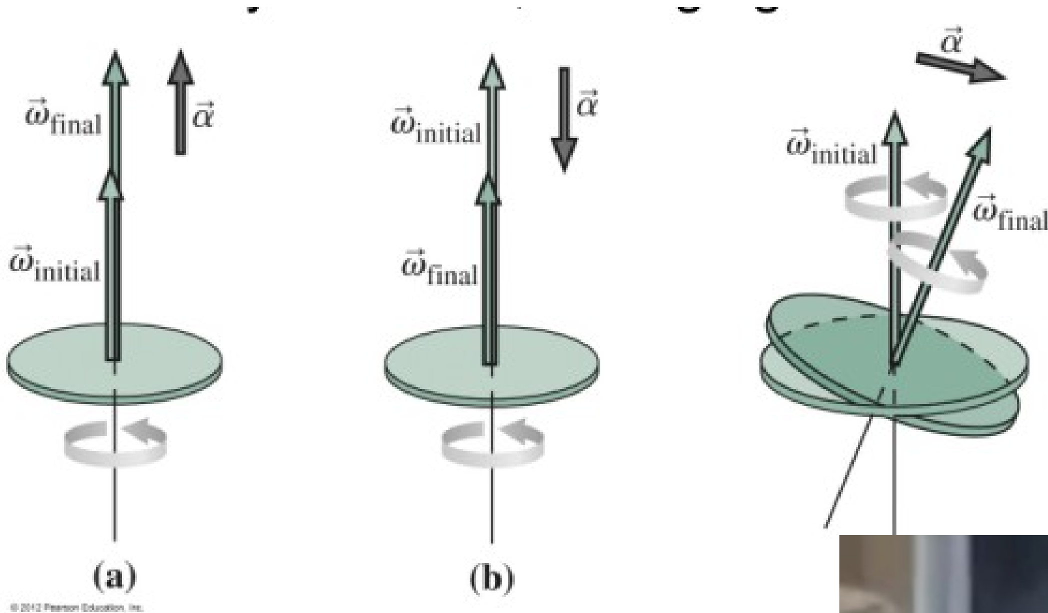

Another movement amplifier!

The MA is smaller than one.
The muscle applies a huge
Force over small distance
(small contraction)

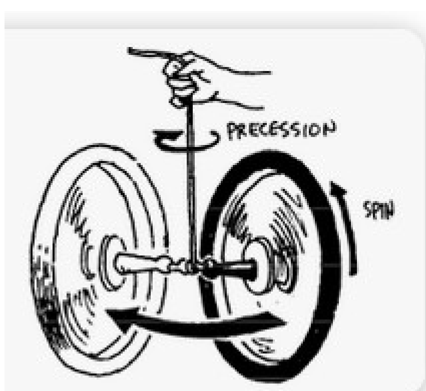
That translates to large
Motion but small force.

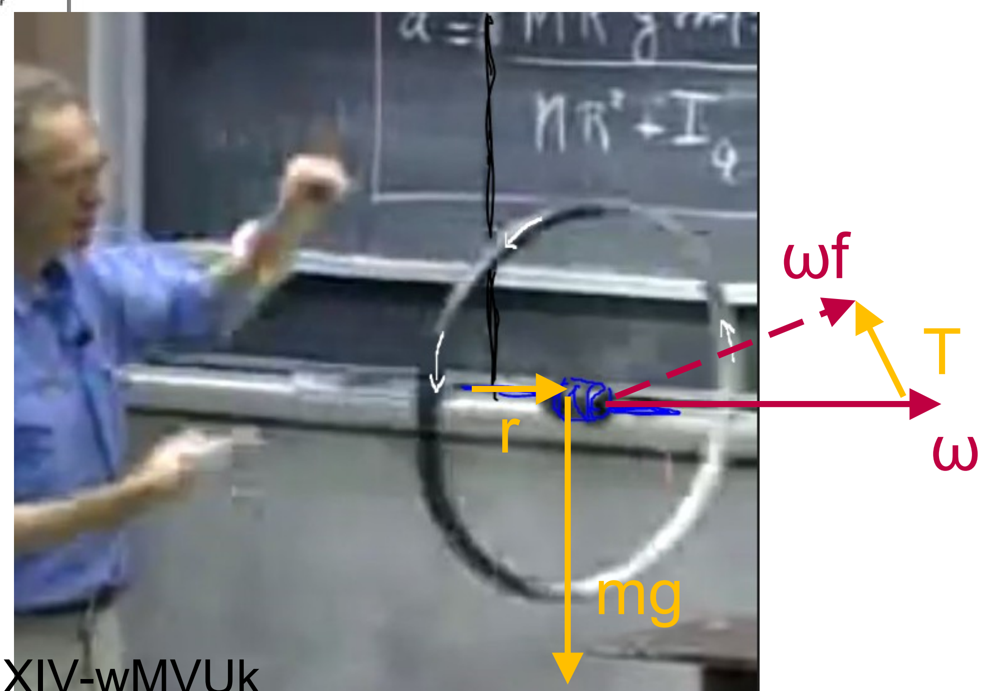
The lower arm bone
Is light. (small load)
But fragile. Easy to break.

# Torque → change magnitude of ω or its direction
## (sum torque = I α )



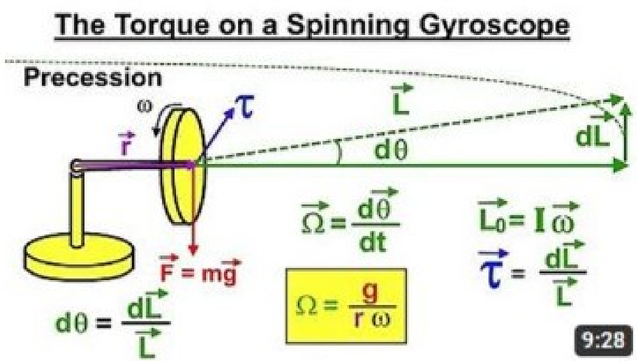

See video Walter Lewin




https://www.youtube.com/watch?v=NeXIV-wMVUk

# Recommended videos for  Physics entusiasts.



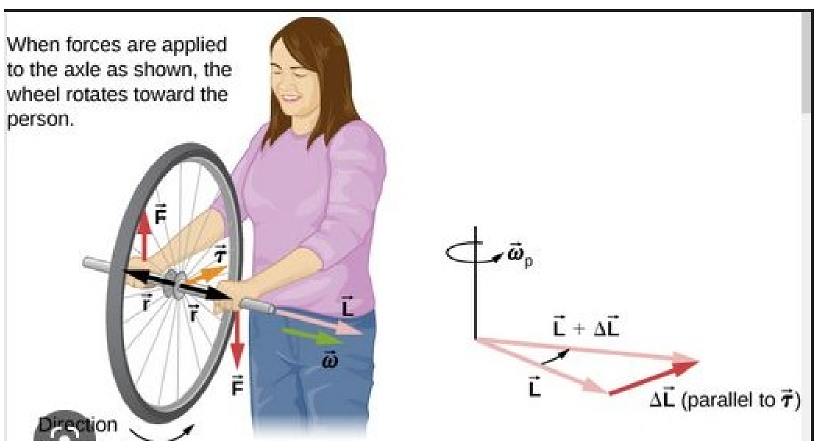

Physics - Mechanics: The Gyroscope (3 of 5) The Torque of a Spinning Gyroscope

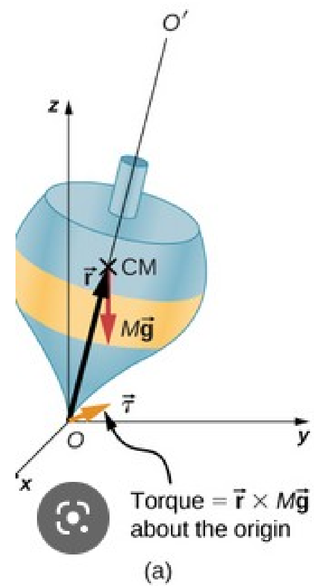
166K views • 6 years ago

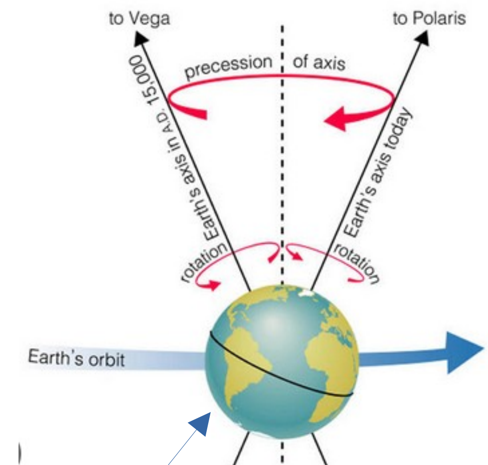
Michel van Biezen

In this video I will explain how **torque** is applied to a **spinning gyroscope**. Next video in this series can be seen at

Angular Momentum | Directional Torque | Precession | Angular Velocity | Moment Arm |...  6 n

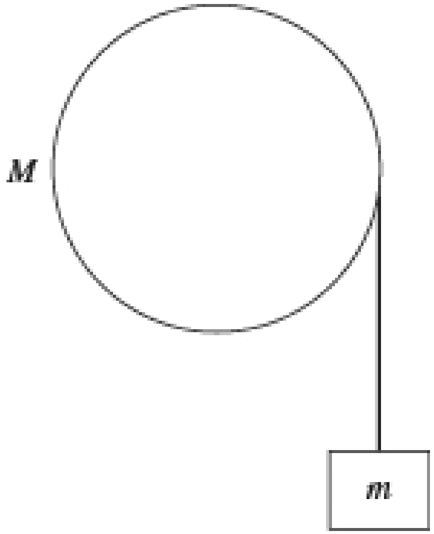



http://hyperphysics.phy-astr.gsu.edu/hbase/Solar/earthprecess.html

3) A cinder block of mass $m = 4.0$ kg is hung from a nylon string that is wrapped around a frictionless pulley having the shape of a cylindrical shell, as shown in the figure. If the cinder block accelerates downward at 4.90 $m/s^2$ when it is released, what is the mass $M$ of the pulley?

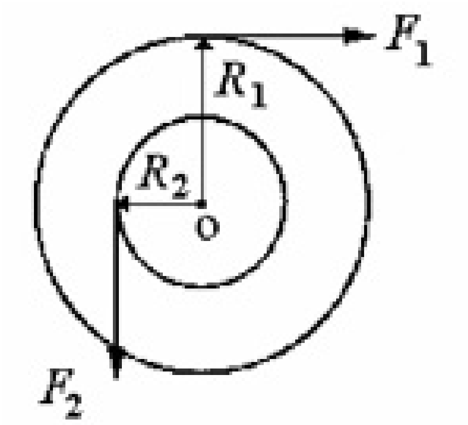


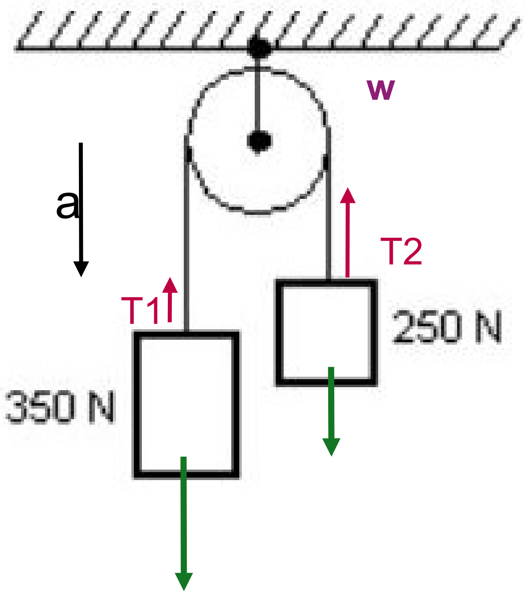

Use the relationship (no slipping of the rope)
Linear acceleration = R (angular acceleration)

Inertia of the pulley is 0.5 MR^2 = Ip

The total inertia of the system is mR^2 + Ip

11) A uniform solid cylinder of mass 10 kg can rotate about a frictionless axle through its center O, as shown in the cross-sectional view in the figure. A rope wrapped around the outer radius $R_1 = 1.0$ m exerts a force of magnitude $F_1 = 5.0$ N to the right. A second rope wrapped around another section of radius $R_2 = 0.50$ m exerts a force of magnitude $F_2 = 6.0$ N downward. What is the angular acceleration of the cylinder?






`1- If the pulley has no mass.

Note: Mg is not equal to the tension
Because down (Mg ) > up (tension)

Say g = 10.
We can have one single system
2 blocks →
350 – 250 = (35 + 25) x a

If we consider just one block→
350 – T = 35 a or
T – 250 = 25 a
**T1 = T2 =T because the pulley has no mass**

**If the pulley has mass →**
**T1 is not equal to T2**
**Because T1 fight against the inertia**
**Of the pulley.**

**Example: M = 5kg**
**Ip = 0.5 M R^2**
**Find a and α**

2- If the pulley has mass and inertia Ip

a = R (α) . Ip = 0.5 $MR^2$ usually

A) If we consider one single system:
(easier). Outside forces = 350N and 250 N
$350R – 250 R = (Ip + 35R^2 + 25R^2)\ \alpha$

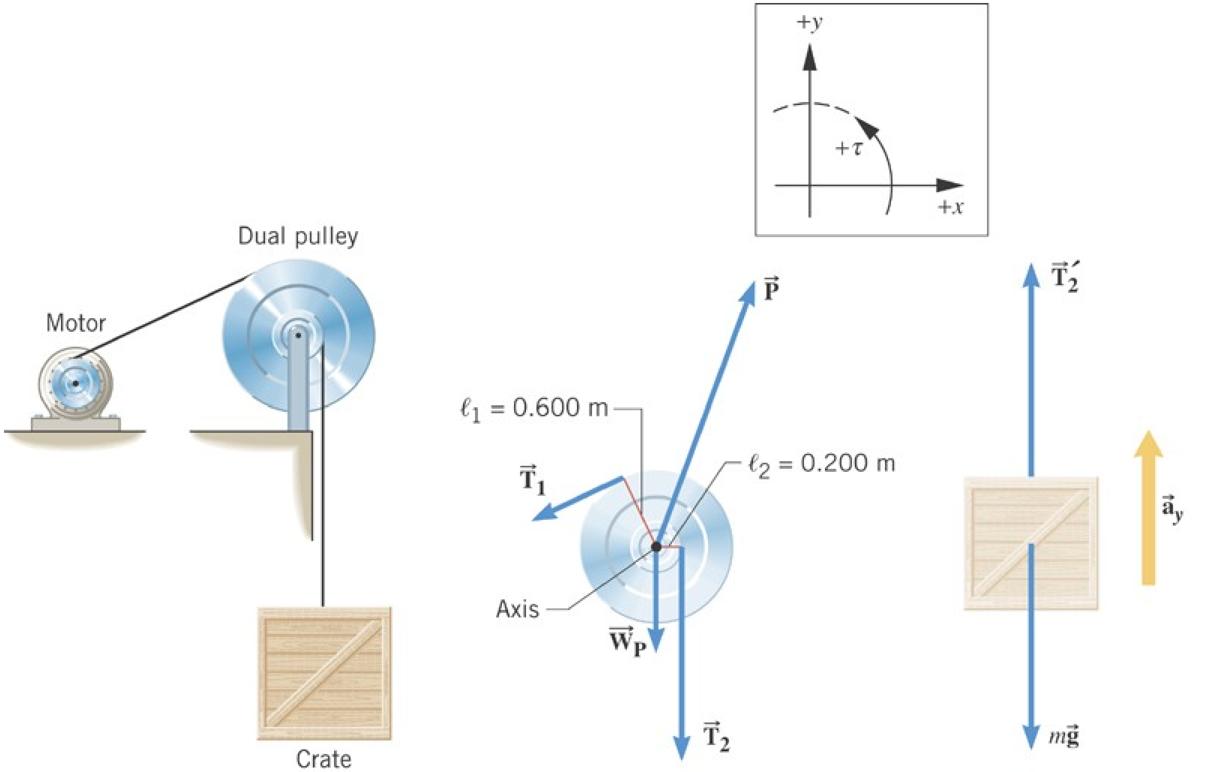
B) To find T1

350 - T1 = 35 x a
With a = R α

C) to find T2
T2 = 250 = 25 a

## *Example 12* Hoisting a Crate

The combined moment of inertia of the dual pulley is 46.0 kg·m$^2$. The crate weighs 4420 N. A tension of 2150 N is maintained in the cable attached to the motor. Find the angular acceleration of the dual pulley.




(*a*) (*b*) Free-body diagram of pulley (*c*) Free-body diagram of crate

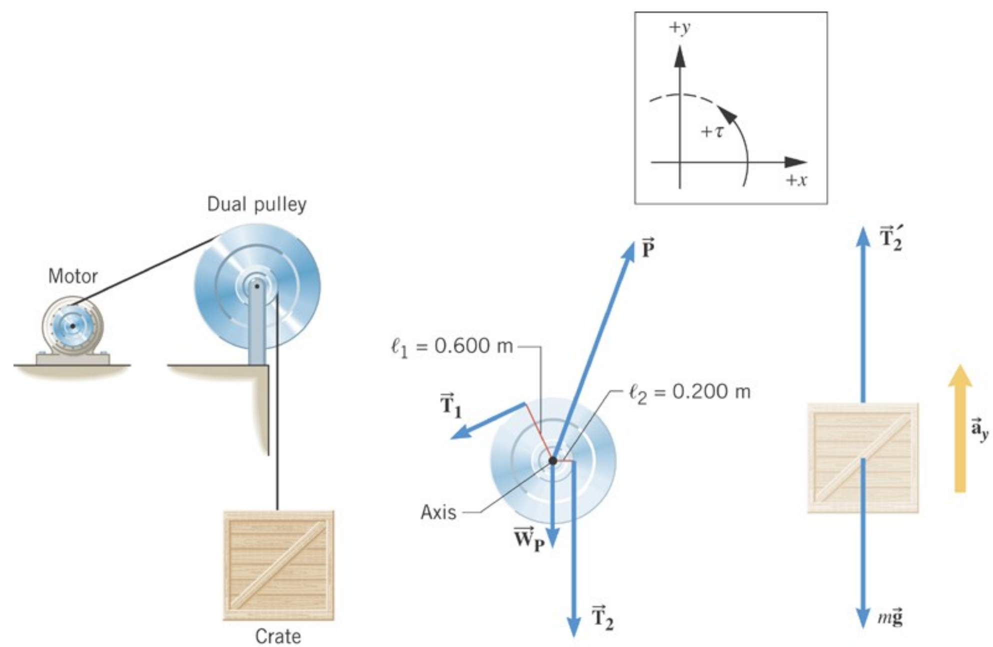
## 9.4 Newton's Second Law for Rotational Motion About a Fixed Axis

equal

$$\sum F_y = T_2' - mg = ma_y \qquad \sum \tau = T_1\ell_1 - T_2\ell_2 = I\alpha$$

$$T_2 = mg + ma_y$$

$$a_y = \ell_2 \alpha$$




(a) (b) Free-body diagram of pulley (c) Free-body diagram of crate

# 9.4 Newton's Second Law for Rotational Motion About a Fixed Axis

$$T_1 \ell_1 - \left(mg + ma_y\right)\ell_2 = I\alpha$$

$$a_y = \ell_2 \alpha$$

$$T_1 \ell_1 - \left(mg + m\ell_2 \alpha\right)\ell_2 = I\alpha$$

$$\alpha = \frac{T_1 \ell_1 - mg\,\ell_2}{I + m\ell_2^2}$$

$$¿\,\frac{(2150\ \text{N})(0.600\ \text{m}) - (451\ \text{kg})(9.80\,m/s^2)(0.200\ \text{m})}{46.0\ \text{kg}\cdot m^2 + (451\ \text{kg})(0.200\ \text{m})^2} = 6.3\ \text{rad}/s^2$$

A wheel that is free to rotate has a radius of 15 centimeters. It has a cord wrapped around its circumference, and a 1.2-kilogram mass is attached to the lower end of the cord. If the mass is released, it falls a distance of 2.5 meters in 3.5 seconds. What is the moment of inertia of the wheel?

Use kinematics to find the acceleration a ($h = 0.5\ a\ t^2$) Then $\alpha = a/R$

$mgR = I\ \alpha$

A disk-shaped 18-kilogram flywheel has a radius of 20 centimeters and is spinning at 150 radians per second. A brake applied to its rim brings it to rest in 12 seconds. How much braking force was used?

Use kinematics to find the acceleration $\alpha$ ($\alpha = (wf - wi) / \text{time}$)

Then force x $R = I\ \alpha$ with $I = MR^2$

# Use the same principle

Masses are connected by a light cord. Find the linear acceleration $a$.

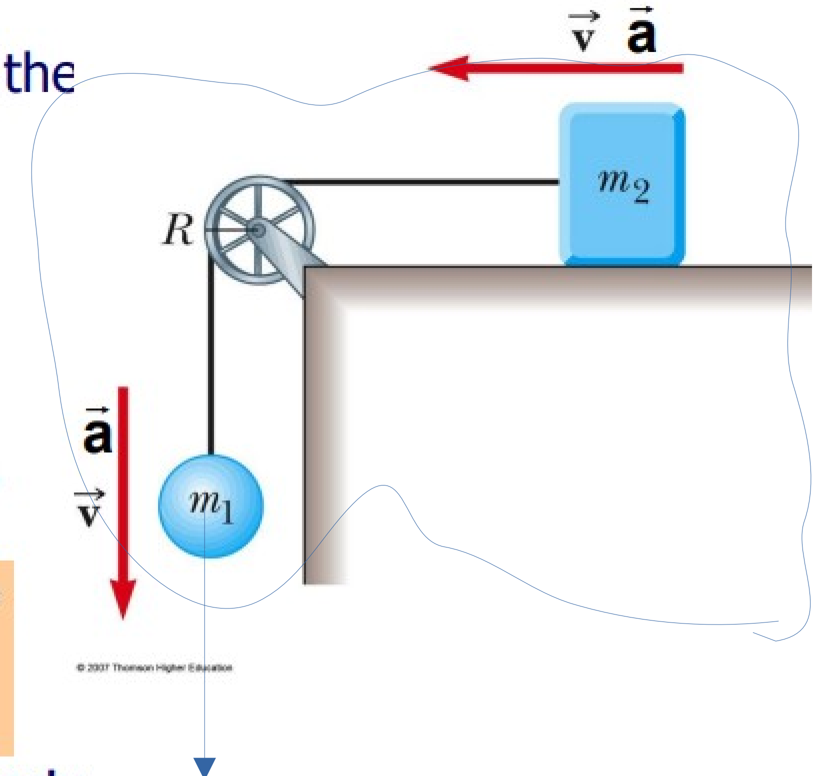
- Use angular momentum approach
- No friction between $m_2$ and table
- Treat block, pulley and sphere as a non-isolated system rotating about pulley axis. As sphere falls, pulley rotates, block slides
- Constraints:

Equal $v$'s and $a$'s for block and sphere

$v = \omega R$ for pulley $\qquad \alpha = d\omega / dt$

$a = \alpha R = dv / dt$

- Ignore internal forces, consider external forces only




Consider the system inside the blue ribbon. The only outside force is m1g
The weight of m2 is balanced by the normal. Does not contribute to motion.
Imagine the masses are close to pulley. Itotal = Ip + m1R$^2$ + m2 R$^2$
(Torque due to m1g) = (total inertia) x (angular acceleration). EASY

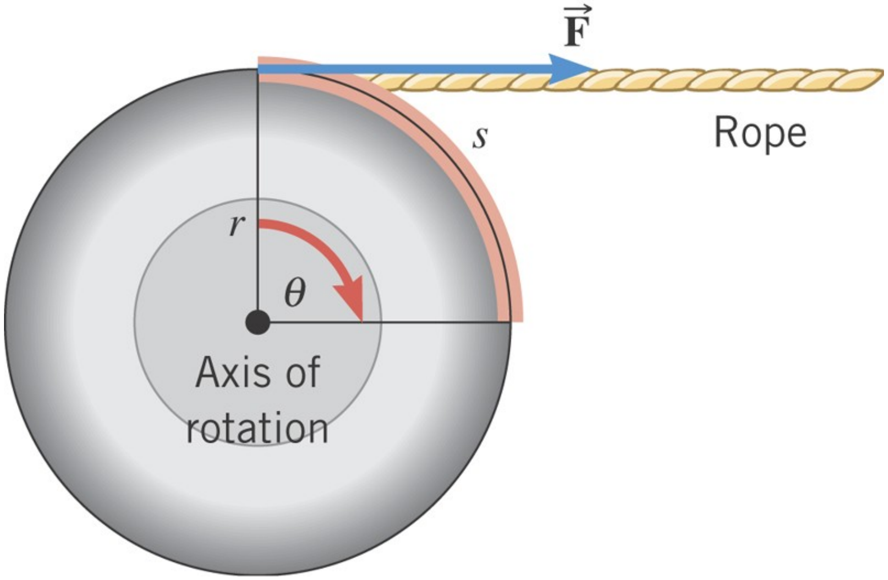
## Rotational Work and Energy

$$s = r\theta$$

$$W = Fs = Fr\,\theta$$

$$\tau = Fr$$

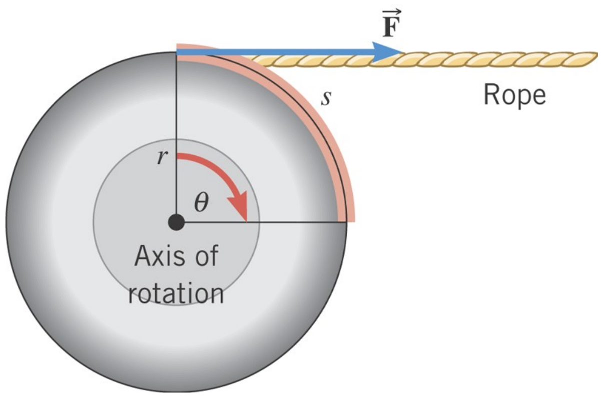
$$W = \tau\theta$$



# DEFINITION OF ROTATIONAL WORK

The rotational work done by a constant torque in turning an object through an angle is

$$W_R = \tau\theta$$




***Requirement:*** The angle must be expressed in radians.

***SI Unit of Rotational Work:*** joule (J)

POWER for rotational motion = Work/time = torque x (angular speed)

Like Power = force x (linear speed)

We can explain gears with conservation of energy
Work in = work out or power in = power out
small angle x big torque = large angle x small torque
small speed x big torque = large speed x small torque

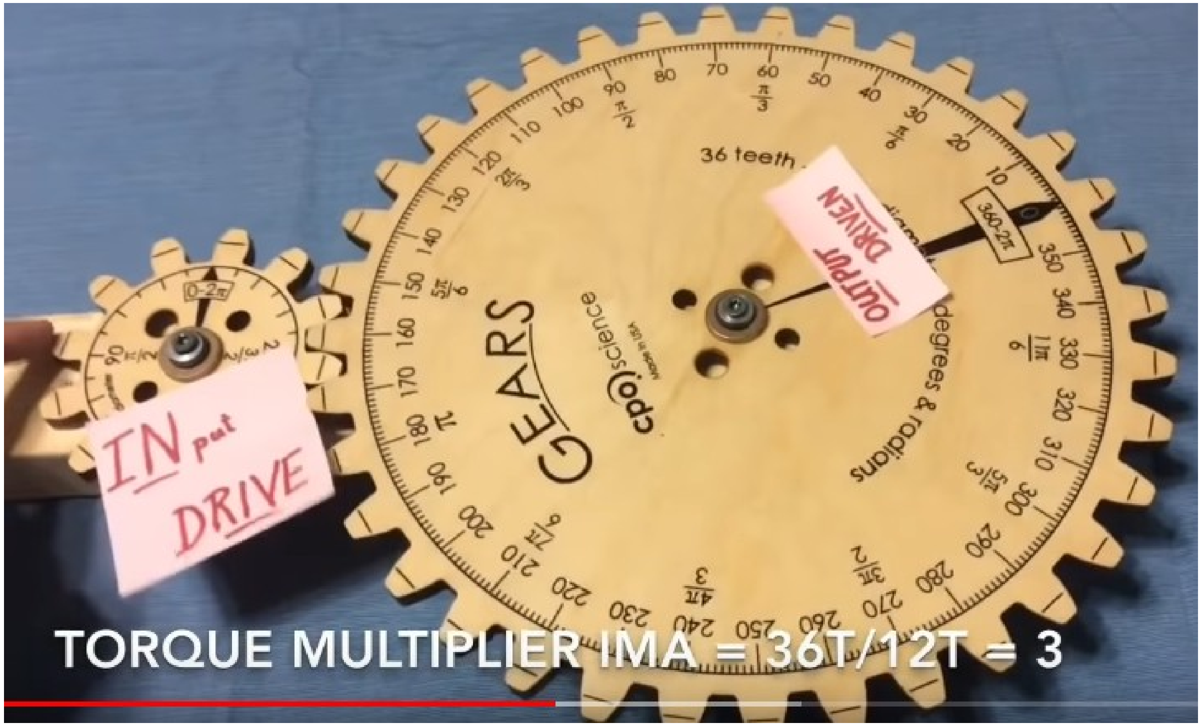

This is a speed Multiplier.

The MA is smaller than 1

Torque1 x angle1 =
Torque2 x angle 2

torque2/torque1 =
angle2/angle1=
# of teeth2/ # teeth 1

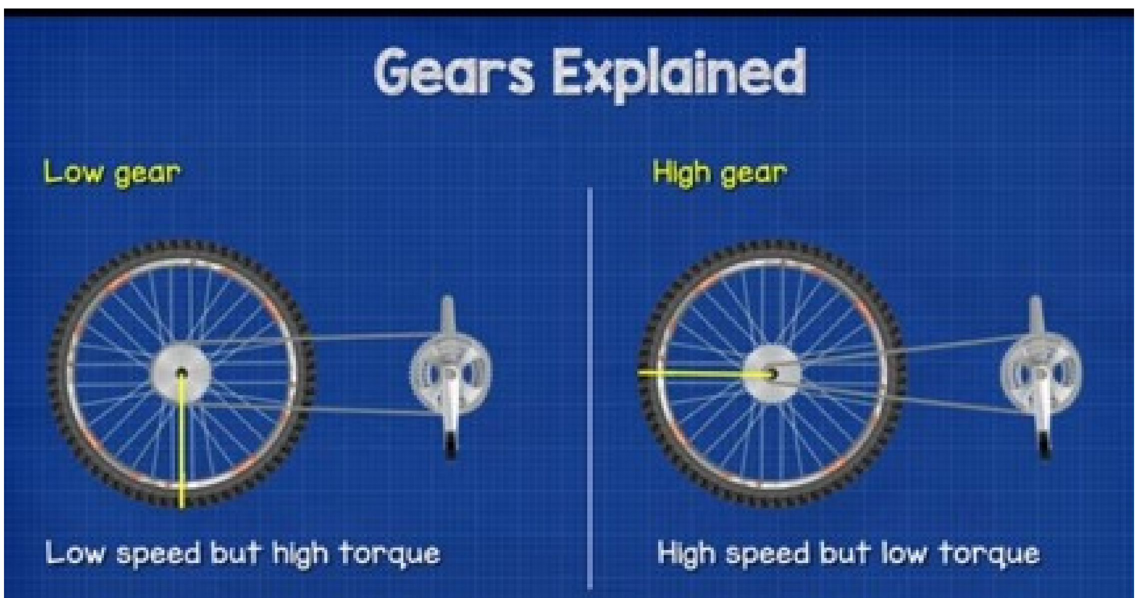
https://www.youtube.com/watch?v=uz436Ixbl-I



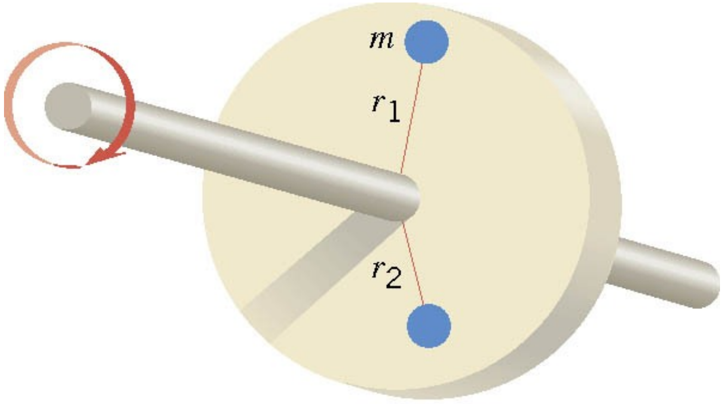

This is a torque multiplier = bike uphill

The MA is larger than 1

F1 x R1x



## 9.5 Rotational Work and Energy




$$KE = \frac{1}{2} m v_T^2 = \frac{1}{2} m r^2 \omega^2$$

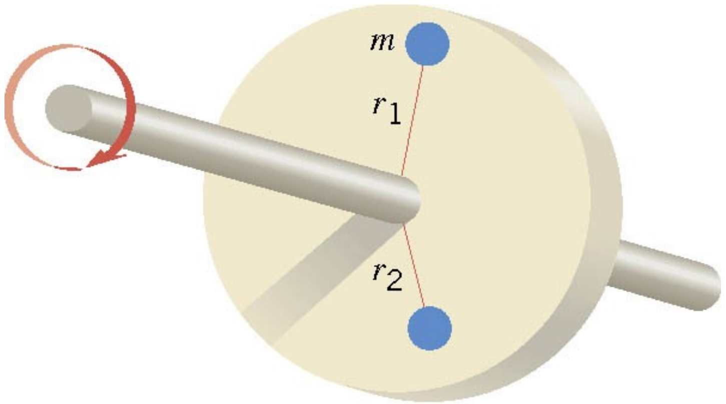
$$v_T = r\omega$$

$$KE = \sum \left( \frac{1}{2} m r^2 \omega^2 \right) = \frac{1}{2} \left( \sum m r^2 \right) \omega^2 = \frac{1}{2} I \omega^2$$

# DEFINITION OF ROTATIONAL KINETIC ENERGY

The rotational kinetic energy of a rigid rotating object is

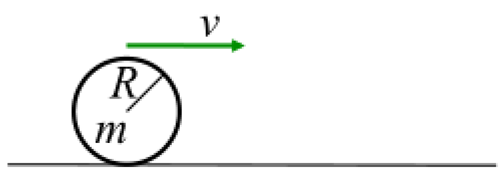
$$KE_R = \frac{1}{2} I\omega^2$$




***Requirement:*** The angular speed must be expressed in rad/s.

***SI Unit of Rotational Kinetic Energy:*** joule (J)

A bowling ball is rolling without slipping at constant speed toward the pins on a lane. What percentage of the ball's total kinetic energy is translational kinetic energy




**a)** 50%

**b)** 71%

**c)** 46%

**d)** 29%

**e)** 33%

Express both KE types with $\omega^2 R^2$

$$KE = \tfrac{1}{2} m v^2, \text{ with } v = \omega R$$

$$KE_{rot} = \tfrac{1}{2} I \omega^2, \text{ with } I = \tfrac{2}{5} m R^2$$

$$\text{fraction} = \frac{KE}{KE + KE_{rot}}$$

b) The moment of inertia is 2/5 (m) $R^2$

10) In a laboratory experiment, a student brings up the rotational speed of a flywheel to 30.0 rpm. She then allows it to slow down on its own, and counts 240 revolutions before the apparatus uniformly comes to a stop. The moment of inertia of the flywheel is 0.0850 $kg \cdot m^2$. What is the magnitude of the retarding torque on the flywheel?

A) 0.000278 $N \cdot m$

B) 0.0000136 $N \cdot m$

C) 0.159 $N \cdot m$

D) 0.0787 $N \cdot m$

E) 0.0425 $N \cdot m$

# Work-KE theorem

Work done by friction = change in KE

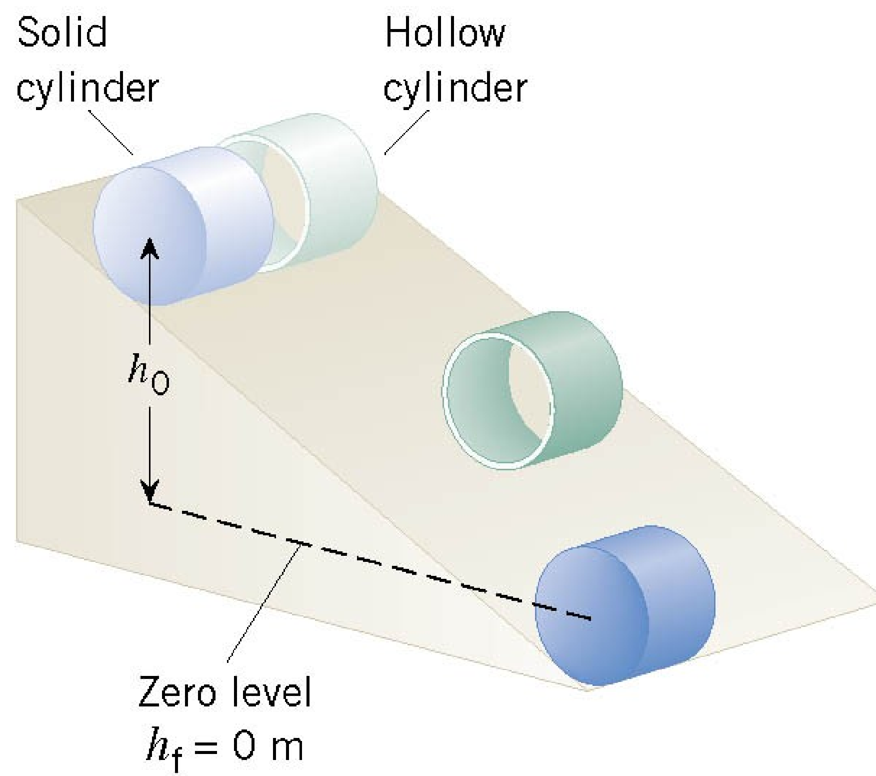
(Torque by friction)x (angular displacement) = 0.5 I (wf^2 – wi^2)

Wf = 0 (it stops). Convert to radians

## *Example 13* Rolling Cylinders

A thin-walled hollow cylinder (mass = $m_h$, radius = $r_h$) and a solid cylinder (mass = $m_s$, radius = $r_s$) start from rest at the top of an incline.

Determine which cylinder has the greatest translational speed upon reaching the bottom.



## 9.5 Rotational Work and Energy

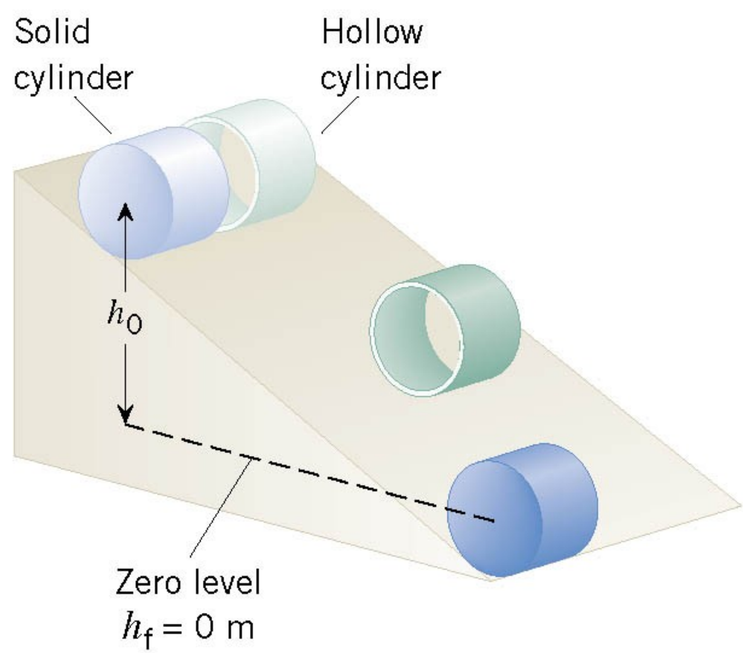
$$E = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + mgh$$

ENERGY CONSERVATION

$$\frac{1}{2}mv_f^2 + \frac{1}{2}I\omega_f^2 + mgh_f = \frac{1}{2}mv_i^2 + \frac{1}{2}I\omega_i^2 + mgh_i$$

$$\frac{1}{2}mv_f^2 + \frac{1}{2}I\omega_f^2 = mgh_i$$

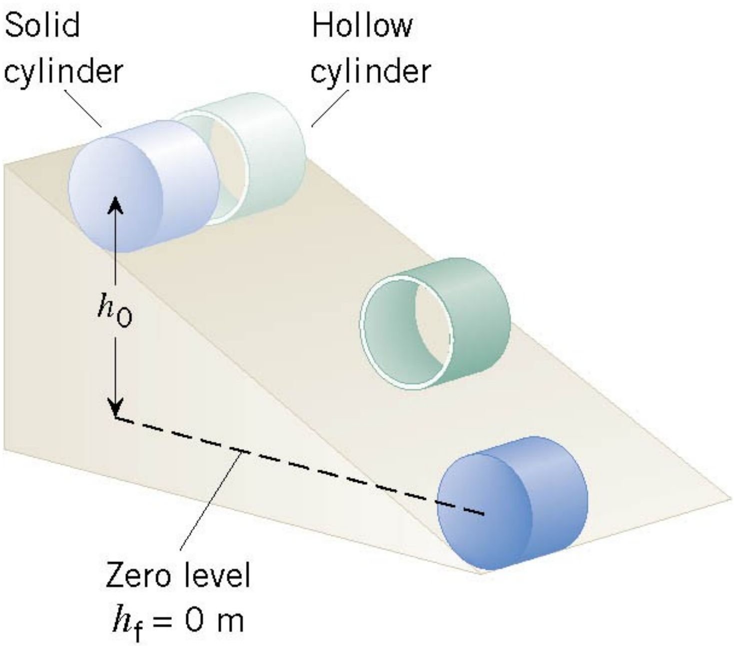
$$\omega_f = v_f / r$$



# 9.5 Rotational Work and Energy

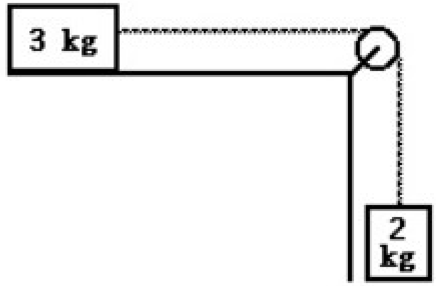
$$\frac{1}{2}mv_f^2+\frac{1}{2}I\,v_f^2/r^2=mgh_i$$

$$v_f=\sqrt{\frac{2mgh_o}{m+I/r^2}}$$




V = sqrt( 2mgh / (m+ coeff x mr$^2$/r$^2$))
V= sqrt (2gh / (1 + coeff) )
coeff = 1 so for a hoop → V = sqrt ( gh)

The cylinder with the smaller moment of inertia will have a greater final translational speed.




9) In Figure 9.2, two blocks, of masses 2 kg and 3 kg, are connected by a light string which passes over a pulley of moment of inertia 0.004 kg $\cdot$ m$^2$ and radius 5 cms. The coefficient of friction for the table top is 0.30. The blocks are released from rest. Using energy methods, one can deduce that after the upper block has moved 0.6 m, its speed is: (see video) 9) ______

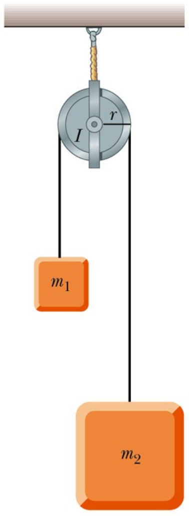
Potential energy of the 2kg – energy lost (friction) = total kinetic energy (with pulley)

2 x g x 0.6 – work friction = linear KE of 2kg + linear KE of 3kg + rotational KE pulley

Linear KE = 0.5 m V^2 and rotation KE = 0.5 I w^2   with V = R w

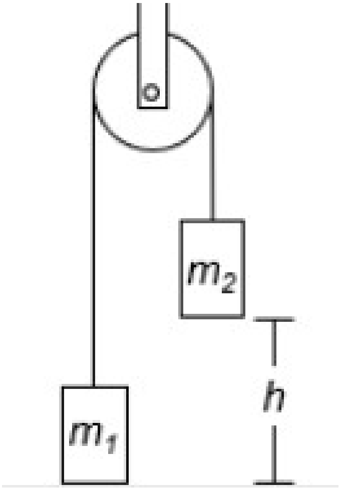
# Blocks and Pulley

- Two blocks having different masses $m_1$ and $m_2$ are connected by a string passing over a pulley. The pulley has a radius R and moment of inertia I about its axis of rotation. The string does not slip on the pulley, and the system is released from rest.





- Find the transnational speeds of the blocks after block 2 descends through a distance h.
- Find the angular speed of the pulley at that time.



Other way: V ^2 = 2 h a and solve for a using 3 equations.

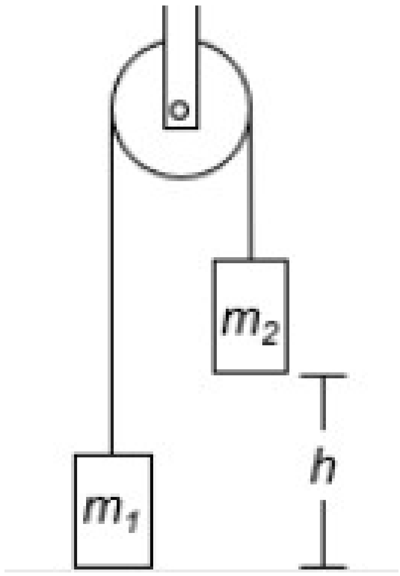
- Find the translational speeds of the blocks after block 2 descends through a distance h.

$$K_{rot,f} + K_{cm,f} + U_f = K_{rot,i} + K_{cm,i} + U_i$$

$$\left(\frac{1}{2}m_1 v_f^{\ 2} + \frac{1}{2}m_2 v_f^{\ 2} + \frac{1}{2}I\omega_f^{\ 2}\right) + (m_1 gh - m_2 gh) = 0 + 0 + 0$$

$$\frac{1}{2}\left(m_1 + m_2 + \frac{I}{R^2}\right)v_f^2 = m_2 gh - m_1 gh$$

$$v_f = \left[\frac{2(m_2 - m_1)gh}{m_1 + m_2 + I/R^2}\right]^{1/2}$$



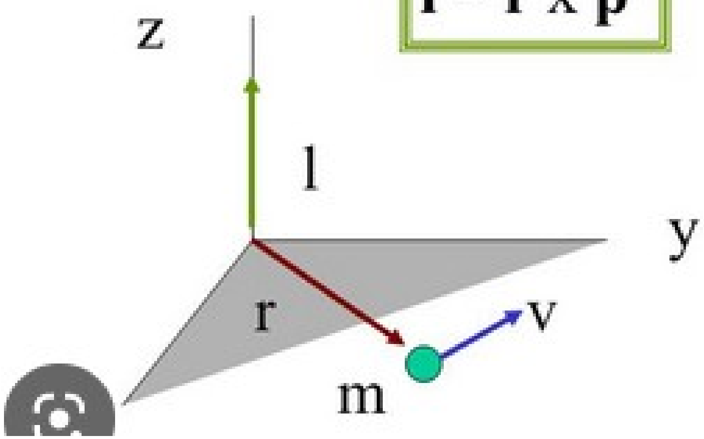

- Find the angular speed of the pulley at that time.

$$\omega_f = \frac{v_f}{R} = \frac{1}{R}\left[\frac{2(m_2 - m_1)gh}{m_1 + m_2 + I/R^2}\right]^{1/2}$$

# Angular Momentum of a Particle

Angular Momentum (l) is the vector cross product of translational momentum and the displacement of that momentum relative to a fixed point :

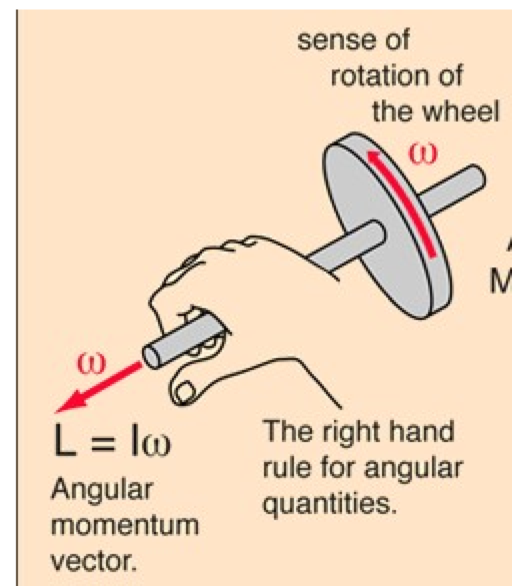
$$\boxed{\mathbf{l} = \mathbf{r} \times \mathbf{p}}$$




Since angular momentum is a vector cross product its direction follows the right hand rule!

The angular momentum is computed relative
To a point. And the angular momentum
Changes with the point.

Unless the solid is rotating about the CM →
L does not depend anymore on the point.



## DEFINITION OF ANGULAR MOMENTUM

The angular momentum *L* of a body rotating about a fixed axis is the product of the body's moment of inertia and its angular velocity with respect to that axis:

$$L = I\omega$$

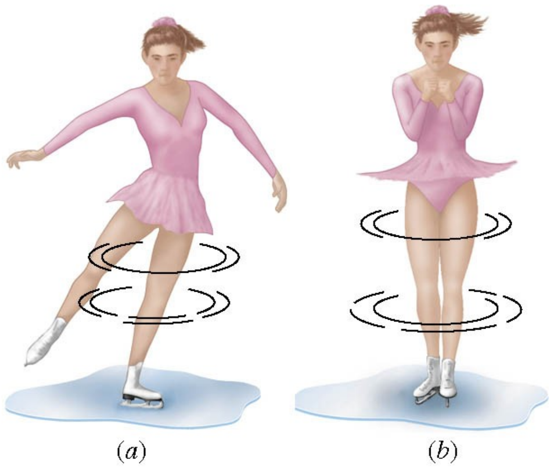
***Requirement:*** The angular speed must be expressed in rad/s.

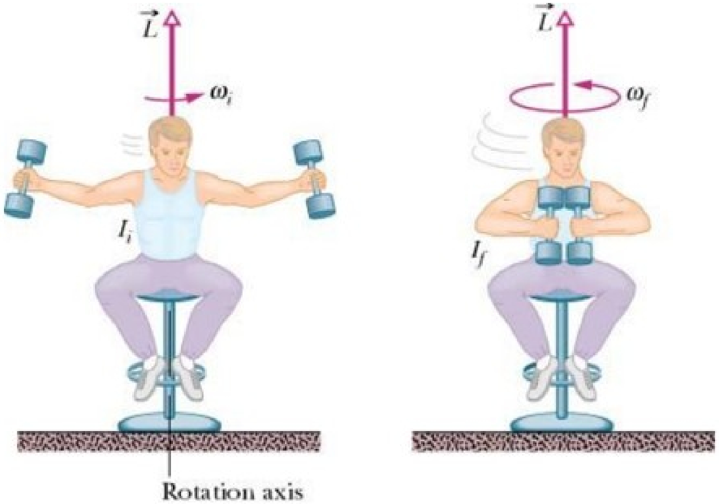
***SI Unit of Angular Momentum:*** kg·m$^2$/s

This is a special case of a symmetric object. The axis goes through the CM
The angular momentum is an intrinsic property of the system.
Otherwise, the angular momentum depends on a reference point/ (point of origin)

# PRINCIPLE OF CONSERVATION OFANGULAR MOMENTUM

The angular momentum of a system remains constant (is conserved) if the net external torque acting on the system is zero.




Isolated System

## *Conceptual Example 14* A Spinning Skater

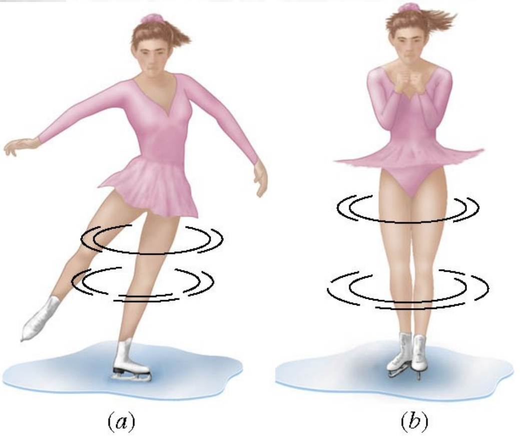
An ice skater is spinning with both arms and a leg outstretched. She pulls her arms and leg inward and her spinning motion changes dramatically.

Use the principle of conservation of angular momentum to explain how and why her spinning motion changes.

(a) (b)

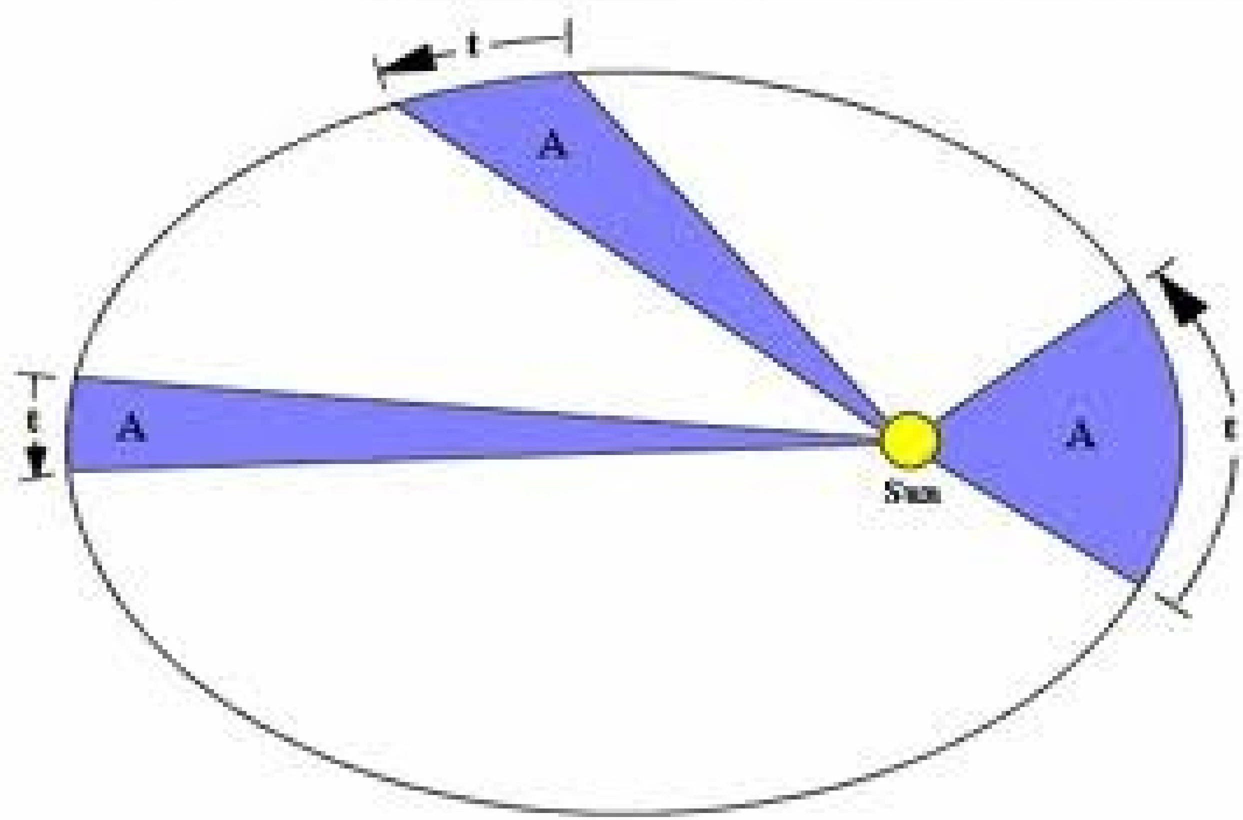
https://www.youtube.com/watch?v=5CZBBbnVmUA head spin

http://www.youtube.com/watch?v=0k276y9kuQQ&feature=related
Simple demo




$$I_1 \, w_1 = I_2 \, w_2$$

Video astro

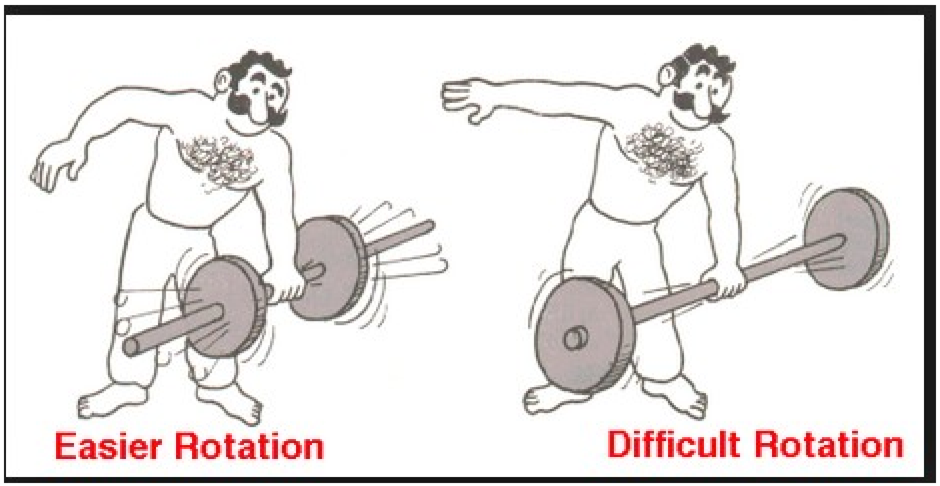



http://highered.mcgraw-hill.com/olcweb/cgi/pluginpop.cgi?it=swf::800::600::/sites/dl/free/0072482621/78778/Kepler_Nav.swf::Keplers%20Second%20Law%20Interactive

Kepler’s second law: the planets sweep the same area during
The same amount of time.
It is a consequence of conservation of angular momentum.

## CONSERVATION OF ANGULAR MOMENTUM

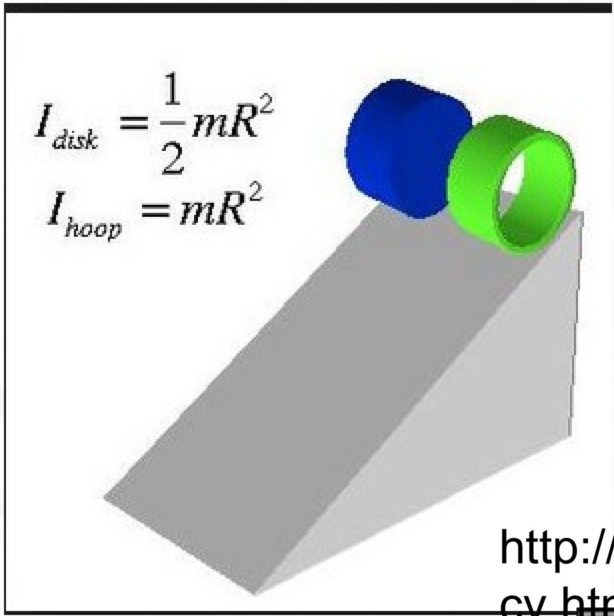
Angular momentum depends on the mass distribution about an axis and On the speed. If the distribution of mass is modified, the speed has to Change so the momentum stays the same.



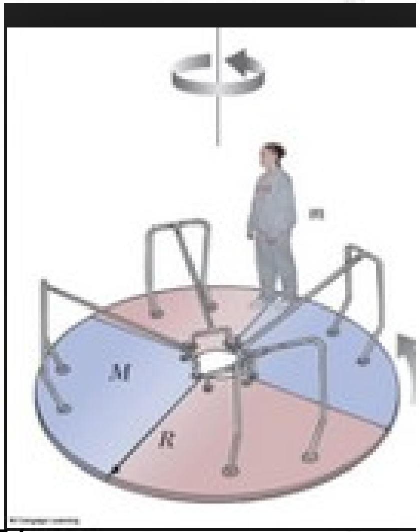

This happens during EQ or tsunami.

2010 a monster quake in Chile shortened the day by 1.26 millionths of A second so the Earth spins (slightly faster)

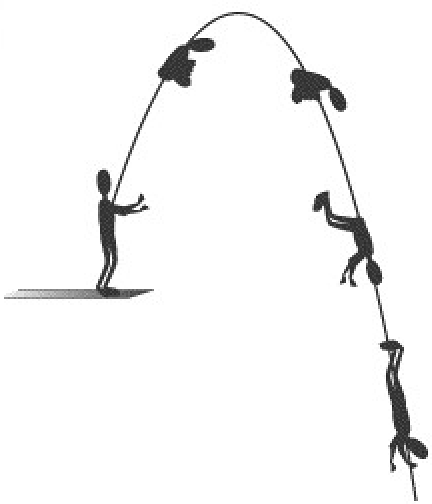
Town of Iloca 168 miles of Santiago. 15 story building https://www.youtube.com/watch?v=tcs93mPn91E

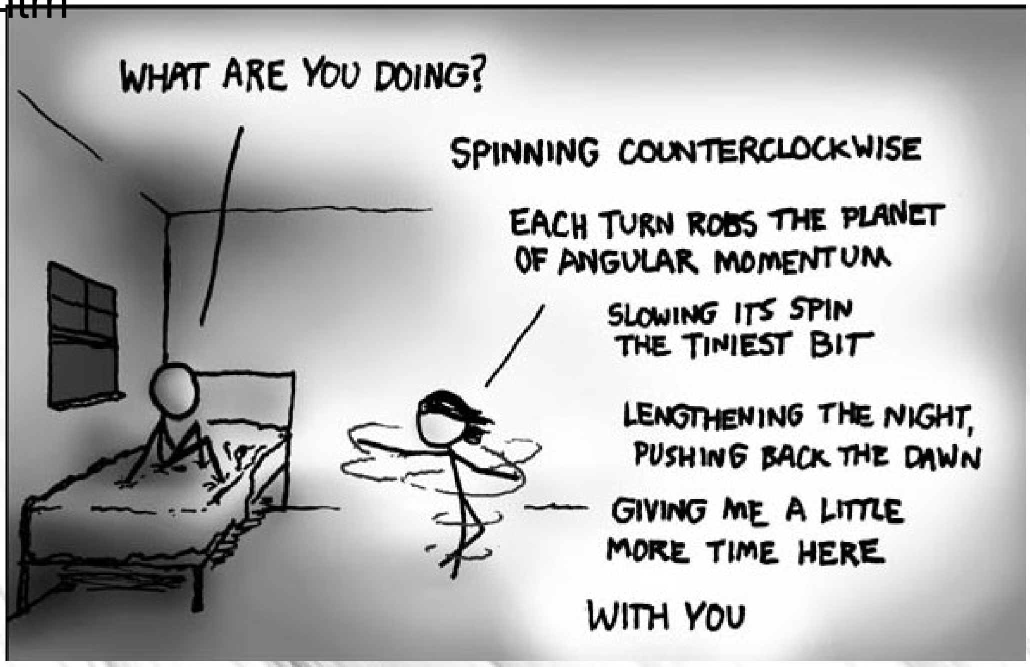


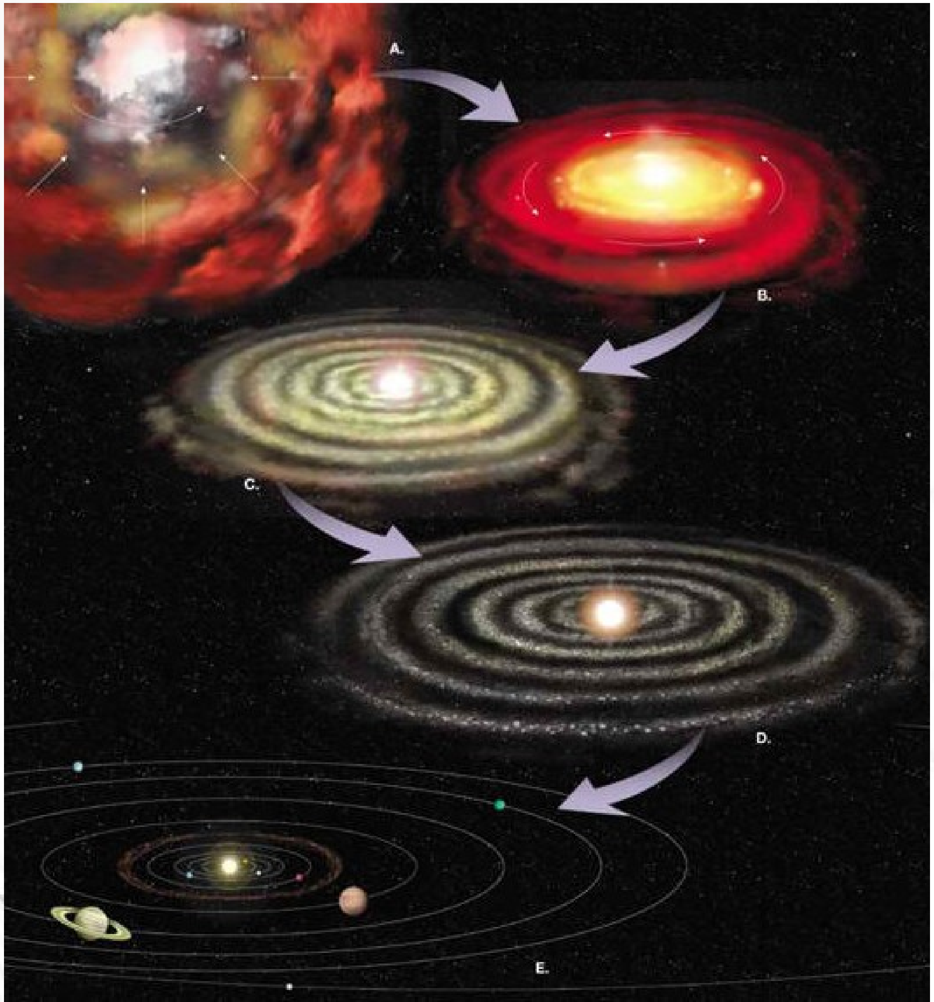

Which one wins ?




http://www.personal.psu.edu/djs75/cv.htm





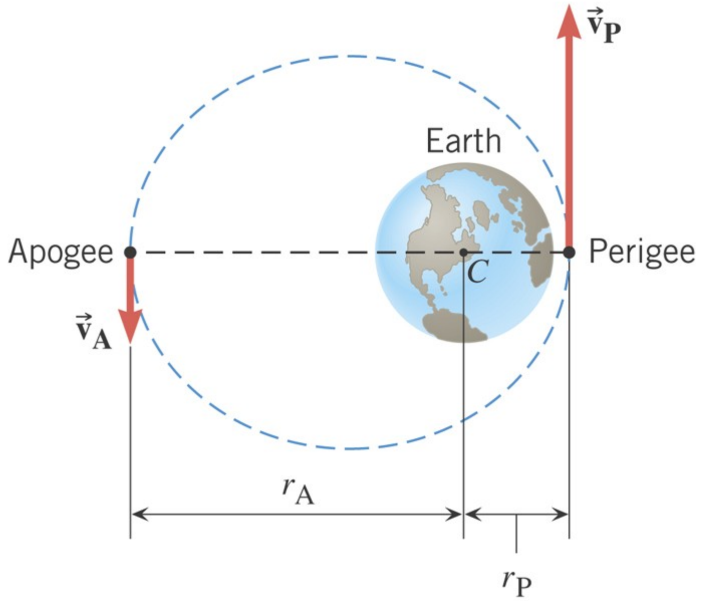


A.
B.
C.
D.
E.

## *Example 15* A Satellite in an Elliptical Orbit

An artificial satellite is placed in an elliptical orbit about the earth. Its point of closest approach is $8.37 \times 10^6$ m from the center of the earth, and its point of greatest distance is $25.1 \times 10^6$ m from the center of the earth.

The speed of the satellite at the perigee is 8450 m/s. Find the speed at the apogee.



# 9.6 Angular Momentum

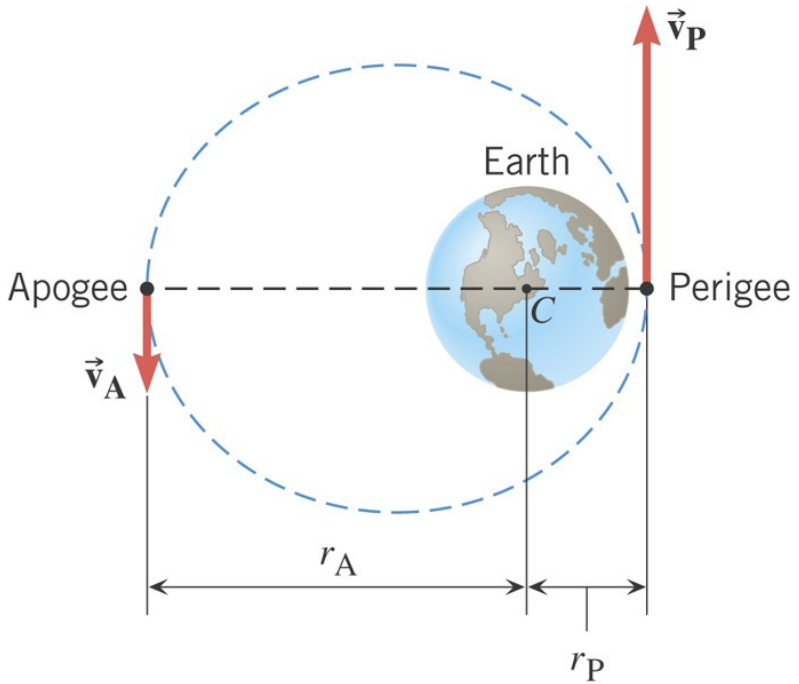
$$L = I\omega$$

angular momentum conservation

$$I_A \omega_A = I_P \omega_P$$

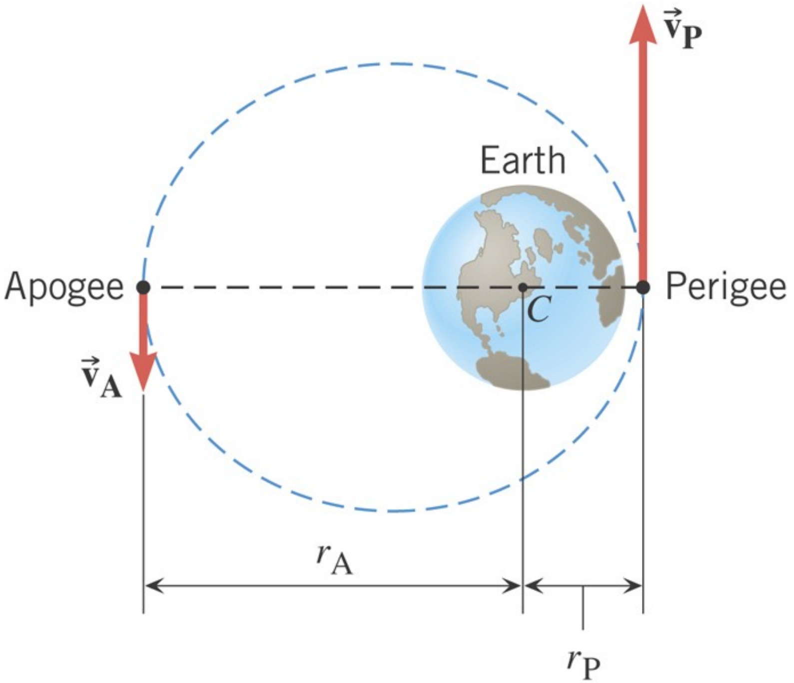
$$I = mr^2 \quad \omega = v/r$$

$$mr_A^2 \frac{v_A}{r_A} = mr_P^2 \frac{v_P}{r_P}$$



# 9.6 Angular Momentum

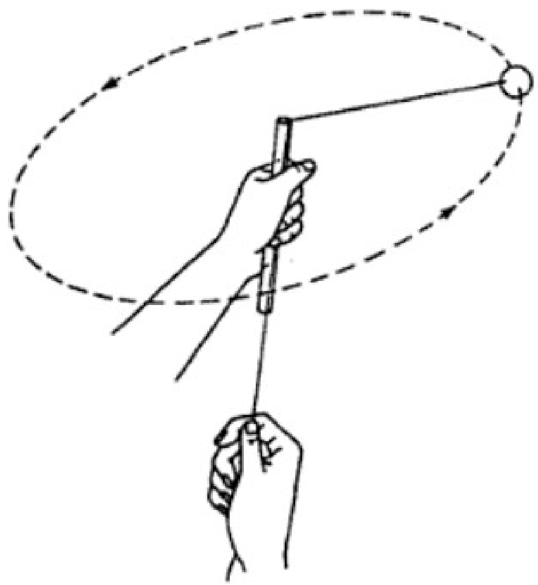
$$mr_A^2\frac{v_A}{r_A}=mr_P^2\frac{v_P}{r_P}$$




$$\boxed{r_A v_A = r_P v_P}$$

$$v_A=\frac{r_P v_P}{r_A}=\frac{\left(8.37\times10^6\text{ m}\right)\left(8450\,m/s\right)}{25.1\times10^6\text{ m}}=2820\,m/s$$

The rotating string in Figure 4-3 is 1.5 meters long, and the ball is going 22 meters per second. How fast will it be going if the string is pulled down until the radius of the circle is reduced to 80 centimeters?



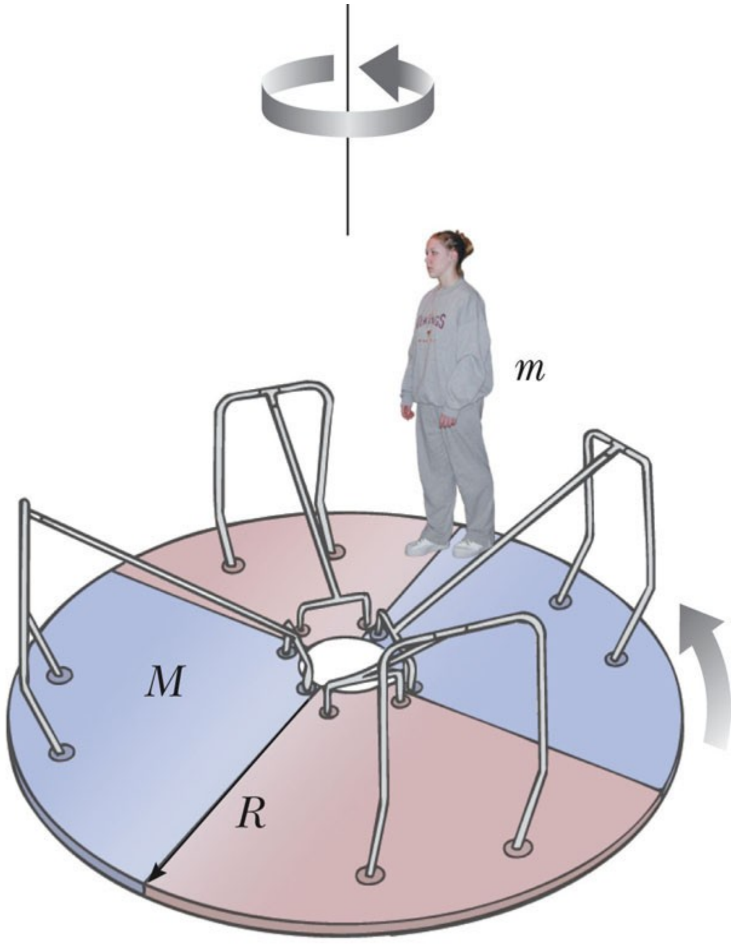
Use R1V1 = R2 V2

A basketball with a mass of 1.2 kilograms and a radius of 18 centimeters is spinning about a vertical axis at 85 radians per second. It is dropped onto the center of a floating 1.60-kilogram ring with a radius of 15 centimeters. What is the angular velocity of the combined objects?

Use I1 w1 = (I1+I2)w2

# The Merry-Go-Round

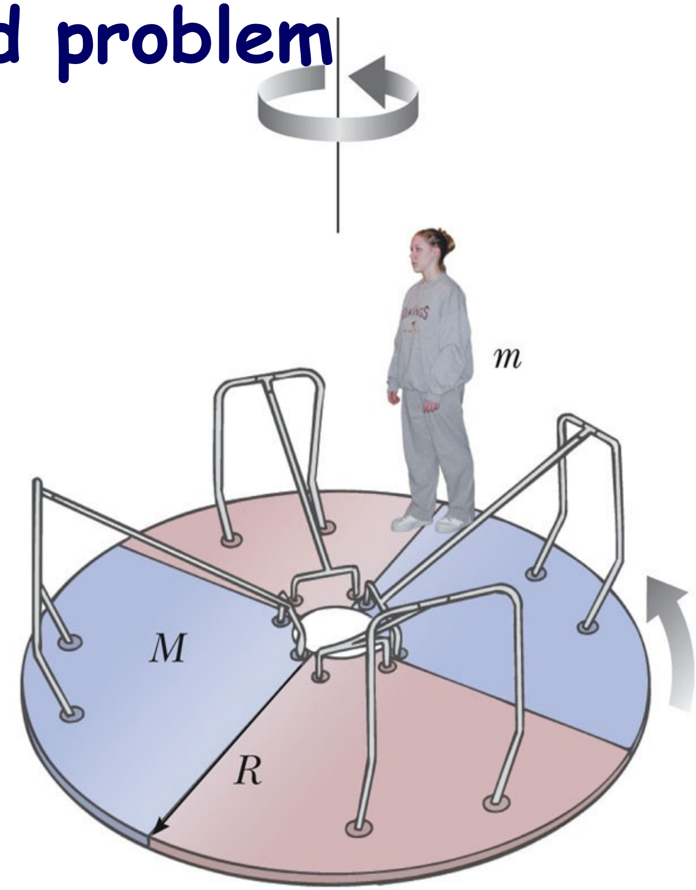
- The moment of inertia of the system = the moment of inertia of the platform plus the moment of inertia of the person.
  - Assume the person can be treated as a particle
- As the person moves toward the center of the rotating platform the moment of inertia decreases.
- The angular speed must increase since the angular momentum is constant.



# Example: A merry-go-round problem

A 40-kg child running at 4.0 m/s jumps tangentially onto a stationary circular merry-go-round platform whose radius is 2.0 m and whose moment of inertia is 20 kg-m$^2$. There is no friction.

Find the angular velocity of the platform after the child has jumped on.





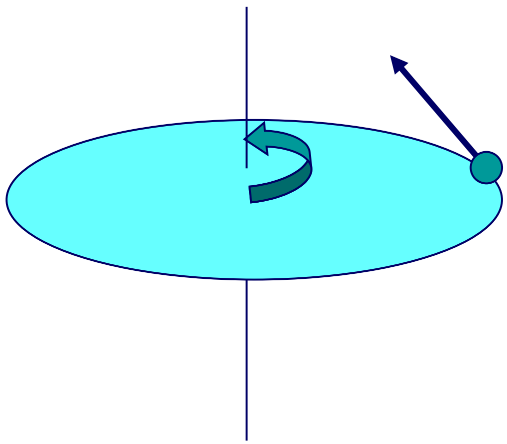
You can use
Momentum before = momentum after
Its a collision.

Before Lchild/axis = mV1 R1 (or mR1$^2$ w1 with V = R1w1)

After Lchild + merry /axis = ( mR1$^2$ + I) w2

The system is isolated → momentum is conserved

# Solution: A merry-go-round problem



$$\mathbf{L}_{tot} = \sum I_i \boldsymbol{\omega}_i = \sum I_f \boldsymbol{\omega}_f$$

$$L_i = I_i\,\omega_i = I\omega_0 + m_c v_T r = m_c v_T r$$

$$L_f = I_f\,\omega_f = \left(I + m_c r^2\right)\omega_f$$

$$\left(I + m_c r^2\right)\omega_f = m_c v_T r$$

$$\omega_f = \frac{m_c v_T r}{I + m_c r^2} = \frac{40 \times 4 \times 2}{10 + 40 \times 2^2} = 1.78 \text{ rad/s}$$

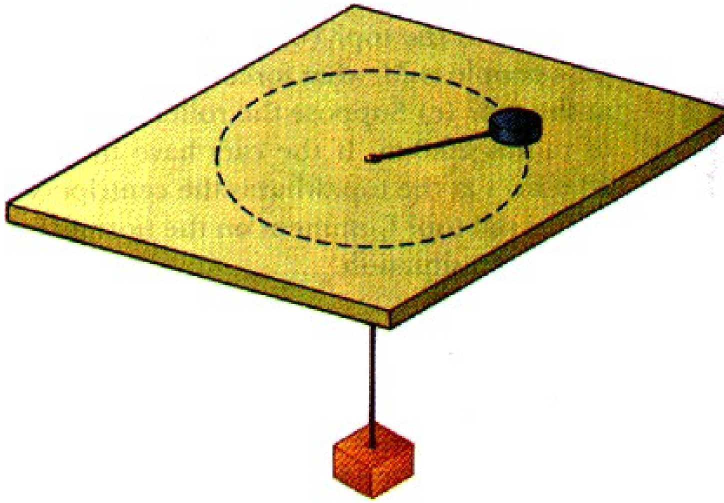
$I = 20$ kg.m$^2$
$V_T = 4.0$ m/s
$m_c = 40$ kg
$r = 2.0$ m
$\omega_0 = 0$

Tipler 10.2

# First Example

A puck of mass m = 0.5 kg is attached to a taut cord passing through a small hole in a frictionless, horizontal surface. The puck is initially orbiting with speed $v_i$ = 2 m/s in a circle of radius

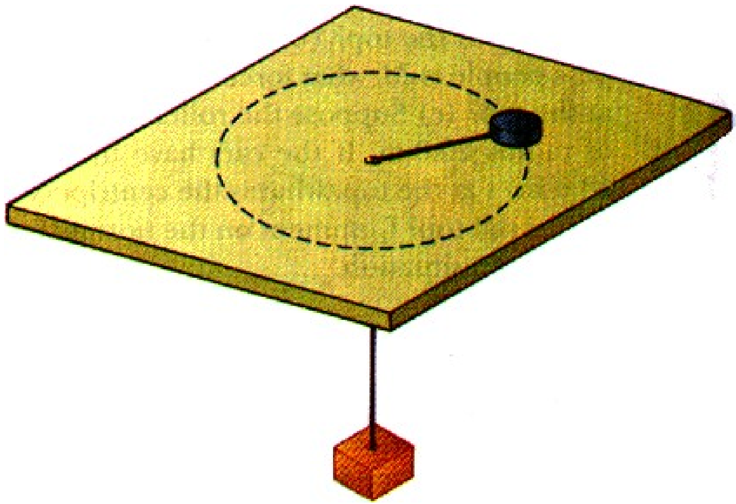
- $r_i$ = 0.2 m. The cord is then slowly pulled from below, decreasing the radius of the circle to r = 0.1 m.
- What is the puck's speed at the smaller radius?
- Find the tension in the cord at the smaller radius.



The angular momentum is conserved because there is
No external torque relative to center.
The torque from pull=0. → R1 V1 = R2 V2
the centripetal force the tension = $mV^2/R$ (slow means T = Fc)

# Angular Momentum Conservation

- $m = 0.5$ kg, $v_i = 2$ m/s, $r_i = 0.2$ m, $r_f = 0.1$ m, $v_f = ?$
- Isolated system?
- Tension force on m exert zero torque about hole, why?



$$\mathbf{L}_i = \mathbf{L}_f \qquad \mathbf{L} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times (m\mathbf{v})$$

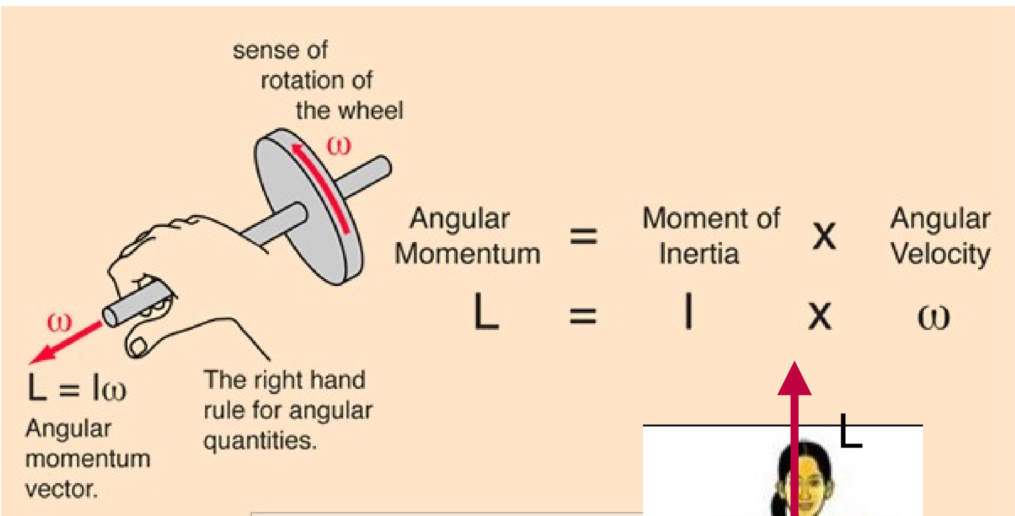
$$L_i = m r_i v_i \sin 90^\circ = m r_i v_i \qquad L_f = m r_f v_f \sin 90^\circ = m r_f v_f$$

$$v_f = \frac{r_i}{r_f} v_i = \frac{0.2}{0.1} 2 = 4\,\text{m/s} \qquad T = m a_c = m \frac{v_f^2}{r_f} = 0.5 \frac{4^2}{0.1} = 80\ \text{N}$$

Tipler 10.6
Textbook 10.12

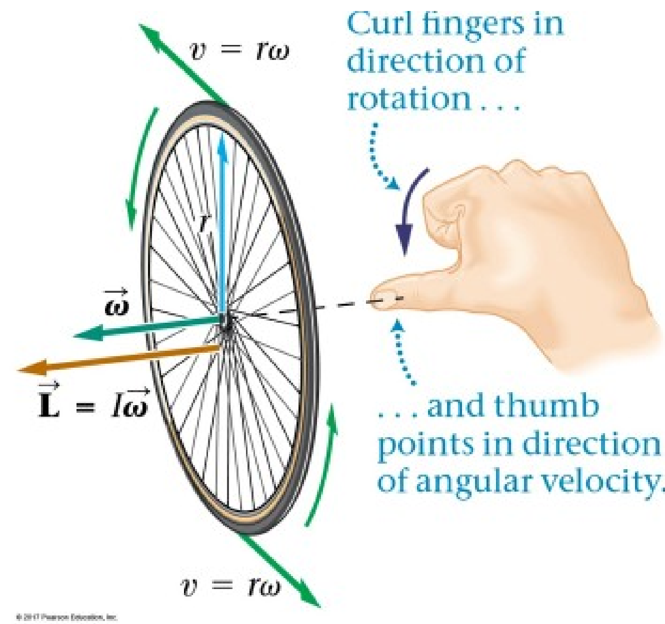
# Angular momentum is a vector

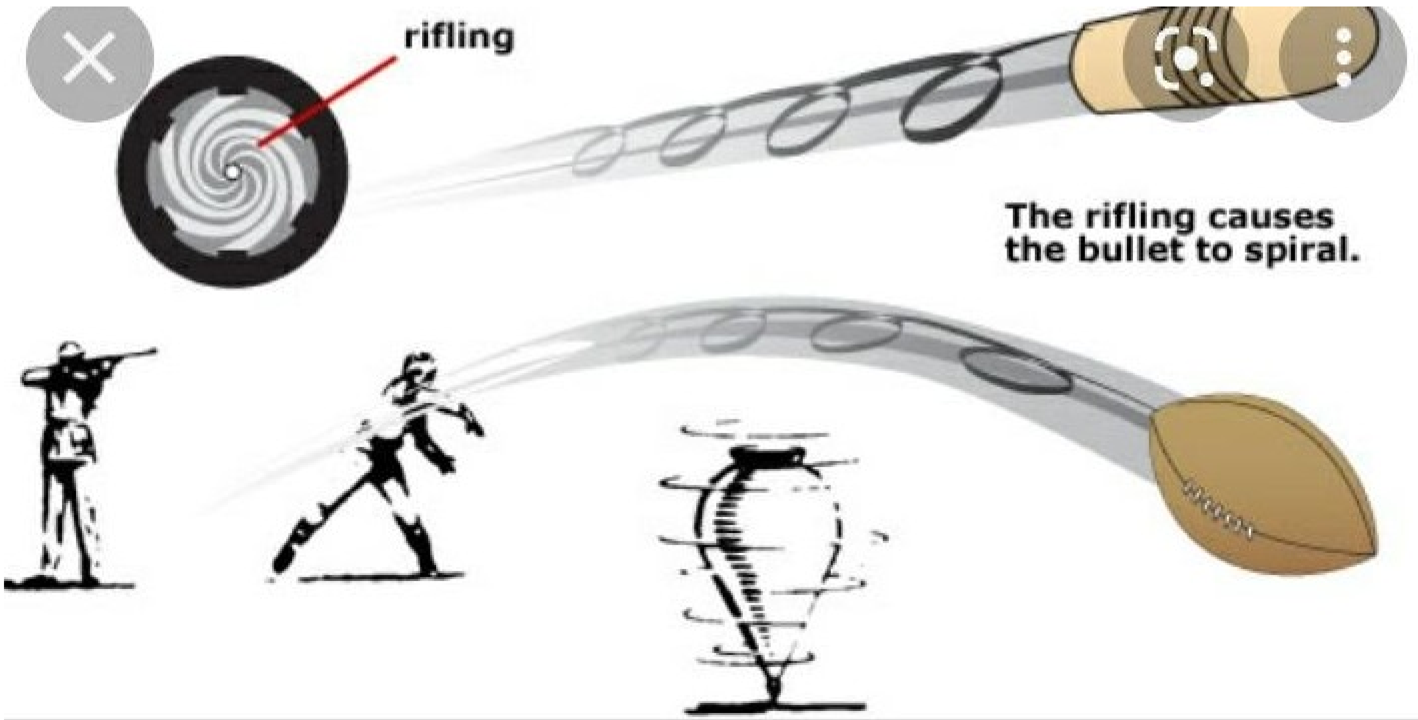
If there is no external torque, the magnitude and direction stays the same. Direction is given by the right hand rule.





Me spinning
(and younger,,)






The vector angular momentum “wants”to point to the same direction
Unless there is an external force. If inertia is large (large mass, large I)
Or if the speed is large, a large torque is needed to destabilize the
Spinning objects. See the video gyroscope in space. →

A gyroscope is Used for orientation

# Definition of Angular Momentum

A particle of mass $m$, traveling with a velocity **v**, has an *angular momentum* relative to any given point in space.

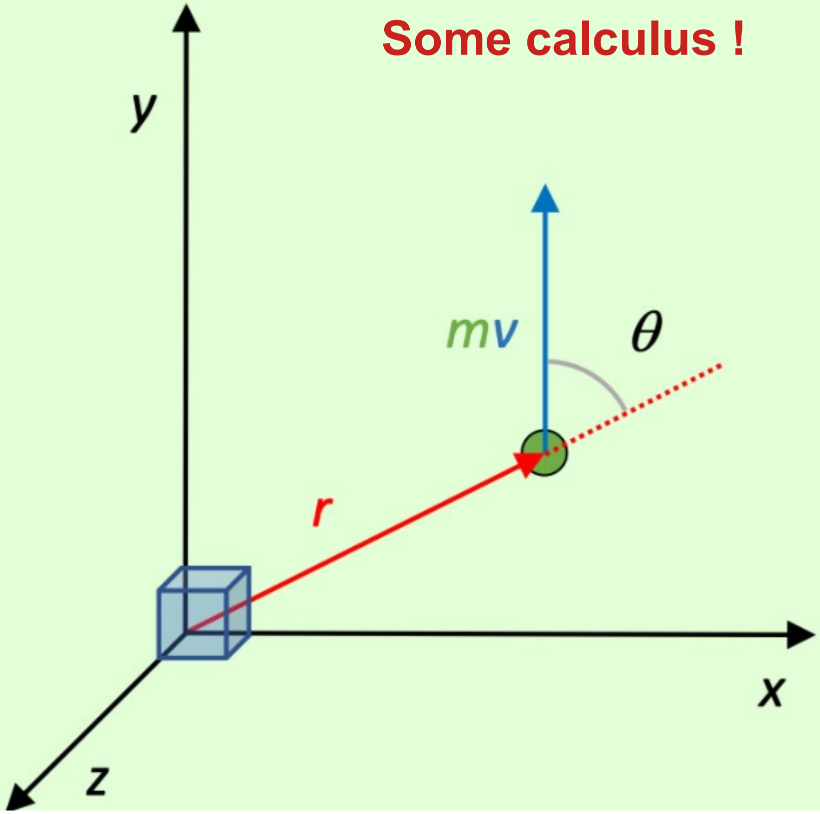
$$\vec{L} = \vec{r} \times \vec{p}$$
$$\vec{L} = \vec{r} \times m\vec{v}$$
$$\vec{L} = rmv\sin\theta$$

We've used the word "angular" to refer to rotational quantities, but it's important to understand that even a mass traveling in a straight line has an angular momentum relative a a given point in space.

Under what circumstances is the angular momentum of a particle o?

Under what circumstances is the angular momentum of a particle $rmv$?

**Some calculus !**




Review cross product
Like

Torque = R X F

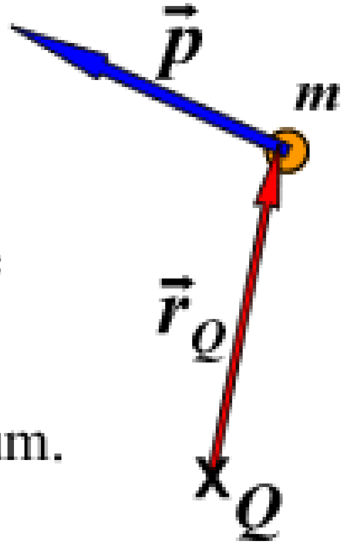
Use right hand rule.

# Angular Momentum = Cross product r x p

$$\vec{L}_Q = \vec{r}_Q \times \vec{p}$$

$\vec{r}_Q$ is the position vector of the particle measured from point Q.

$\vec{p} = m\vec{v}$ is the particle's linear momentum.



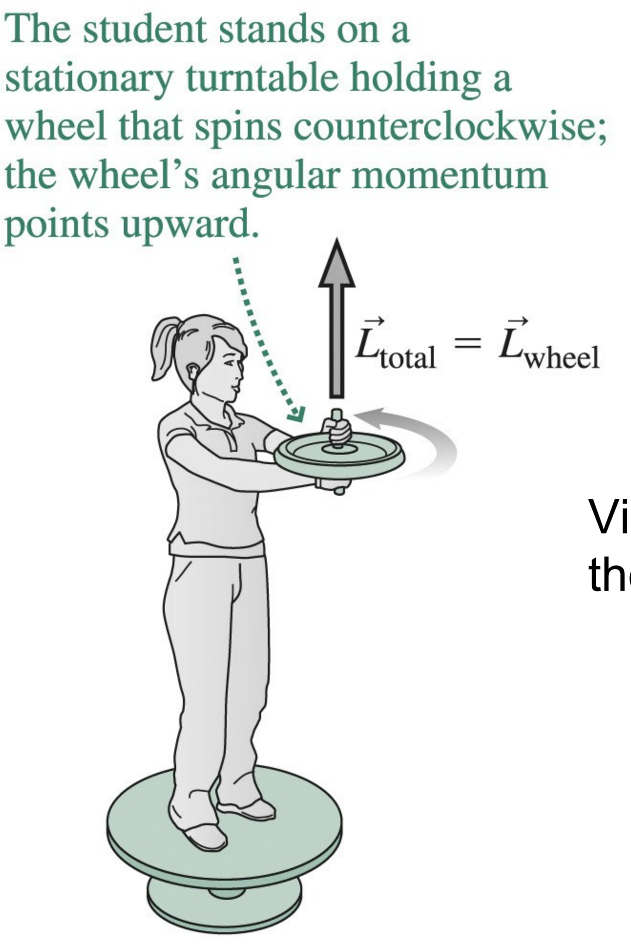

The angular momentum depends on the point of origin.

Except if the axis of rotation goes through the center of mass

Also. Show the right hand angle. L is out of the slide. It is a vector.

Without external torque, momentum stays constant. Magnitude AND DIRECTION.

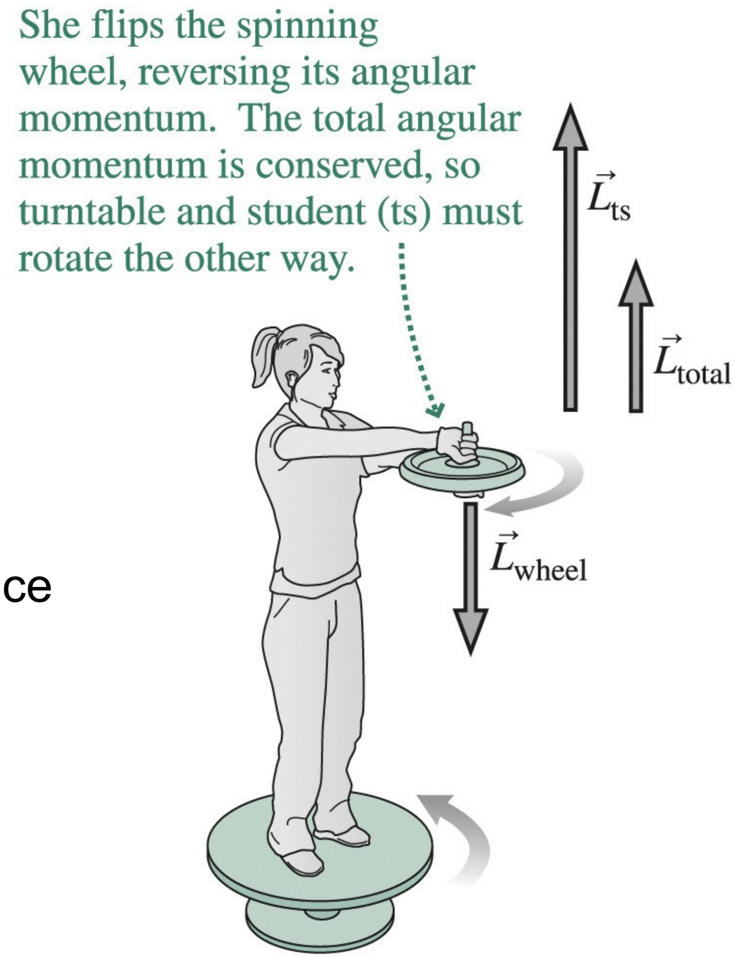
Here before momentum of the system (stool, girl, wheel) is up so without Friction it want to stay up. It the idea of reaction wheels (space, robotics) → managing momentum.




(a)



Video science theater




(b)

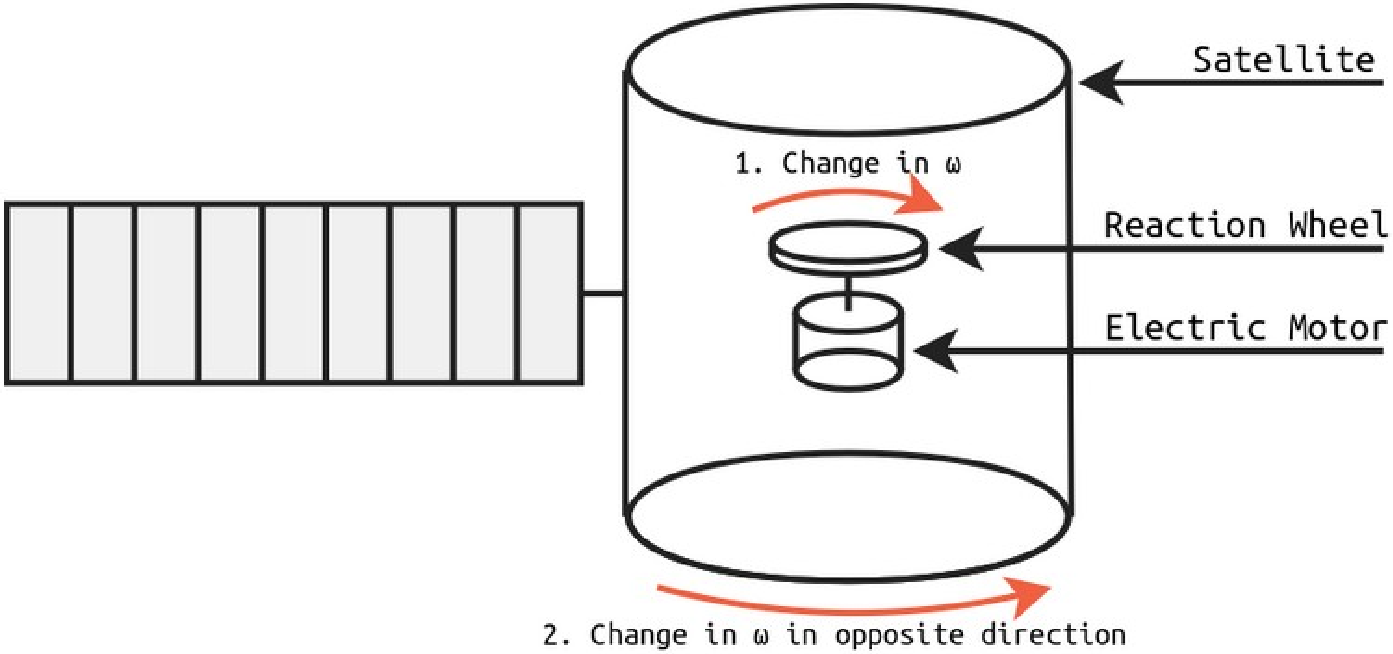
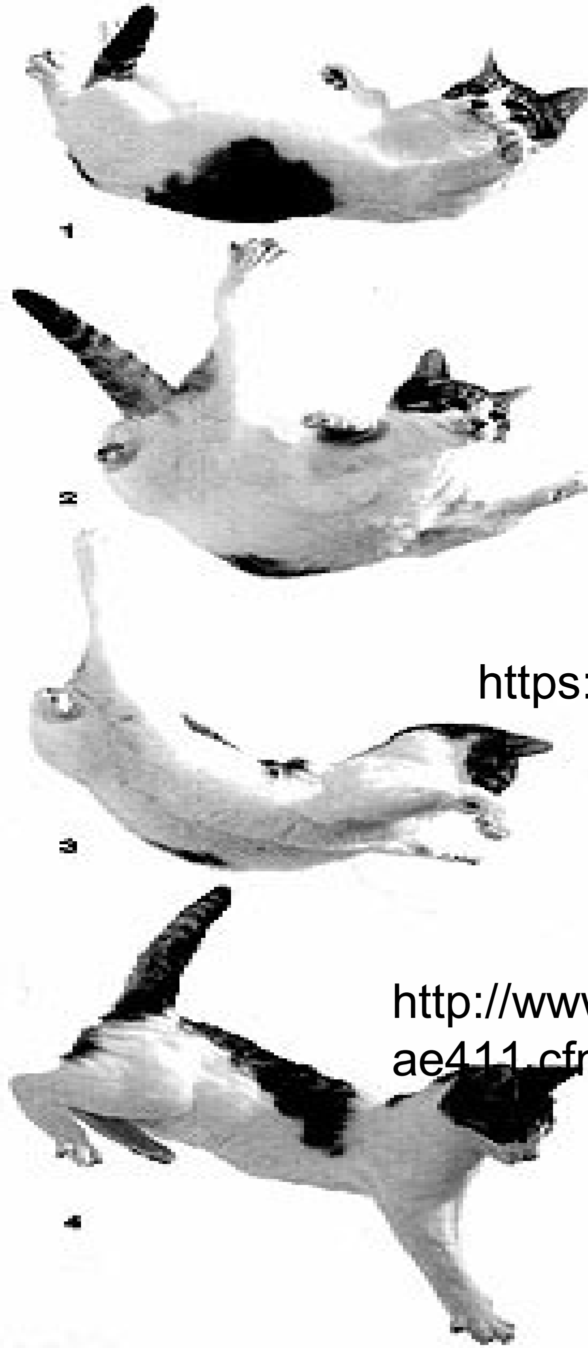


*Diagram of a satellite with reaction wheels*

https://charleslabs.fr/en/project-Reaction+Wheel+Attitude+Control

https://learn.parallax.com/tutorials/robot/elev-8/understanding-physics-multirotor-flight/rotation-torque-and-angular-momentum



The cat spins the tail in one direction
And the body spins the other way.

This is the conservation of momentum.

Same idea with astronauts.

https://www.youtube.com/watch?v=RtWbpyjJqrU

http://www.physlink.com/education/askexperts/ae411.cfm

# Recall: Linear Momentum and Force

- Linear motion: apply force to a mass
- The force causes the linear momentum to change
- The net force acting on a body is the time rate of change of its linear momentum

$$\mathbf{F}_{\text{net}} = \Sigma\mathbf{F} = m\mathbf{a} = m\frac{d\mathbf{v}}{dt} = \frac{d\mathbf{p}}{dt}$$

$$\mathbf{p} = m\mathbf{v}$$

$$\mathbf{I} = \mathbf{F}_{\text{net}}\Delta t = \Delta\mathbf{p}$$

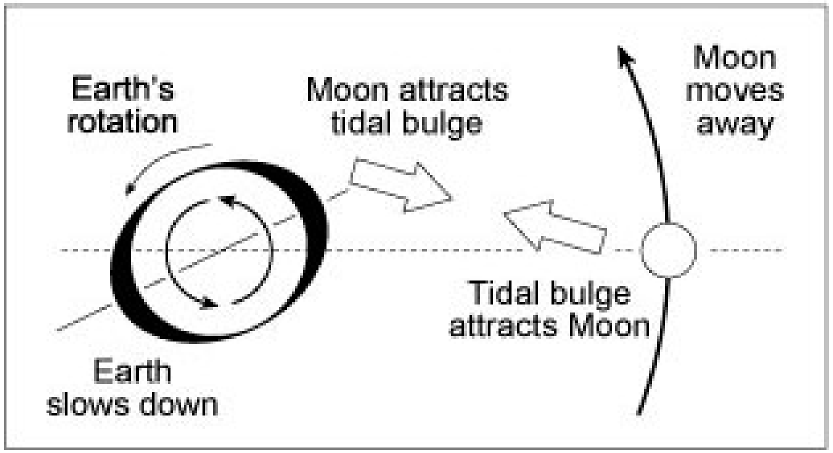
A force applied during a time t → changes the linear momentum
A torque applied during a tome t → changed the angular momentu,

# Angular Momentum and Torque

- Rotational motion: apply torque to a rigid body
- The torque causes the angular momentum to change
- The net torque acting on a body is the time rate of change of its angular momentum

$$\mathbf{F}_{\text{net}} = \Sigma\mathbf{F} = \frac{d\mathbf{p}}{dt} \quad \Rightarrow \quad \boldsymbol{\tau}_{\text{net}} = \Sigma\boldsymbol{\tau} = \frac{d\mathbf{L}}{dt}$$

- $\Sigma\boldsymbol{\tau}$ and $\mathbf{L}$ are to be measured about the same origin
- The origin must not be accelerating (must be an inertial frame)



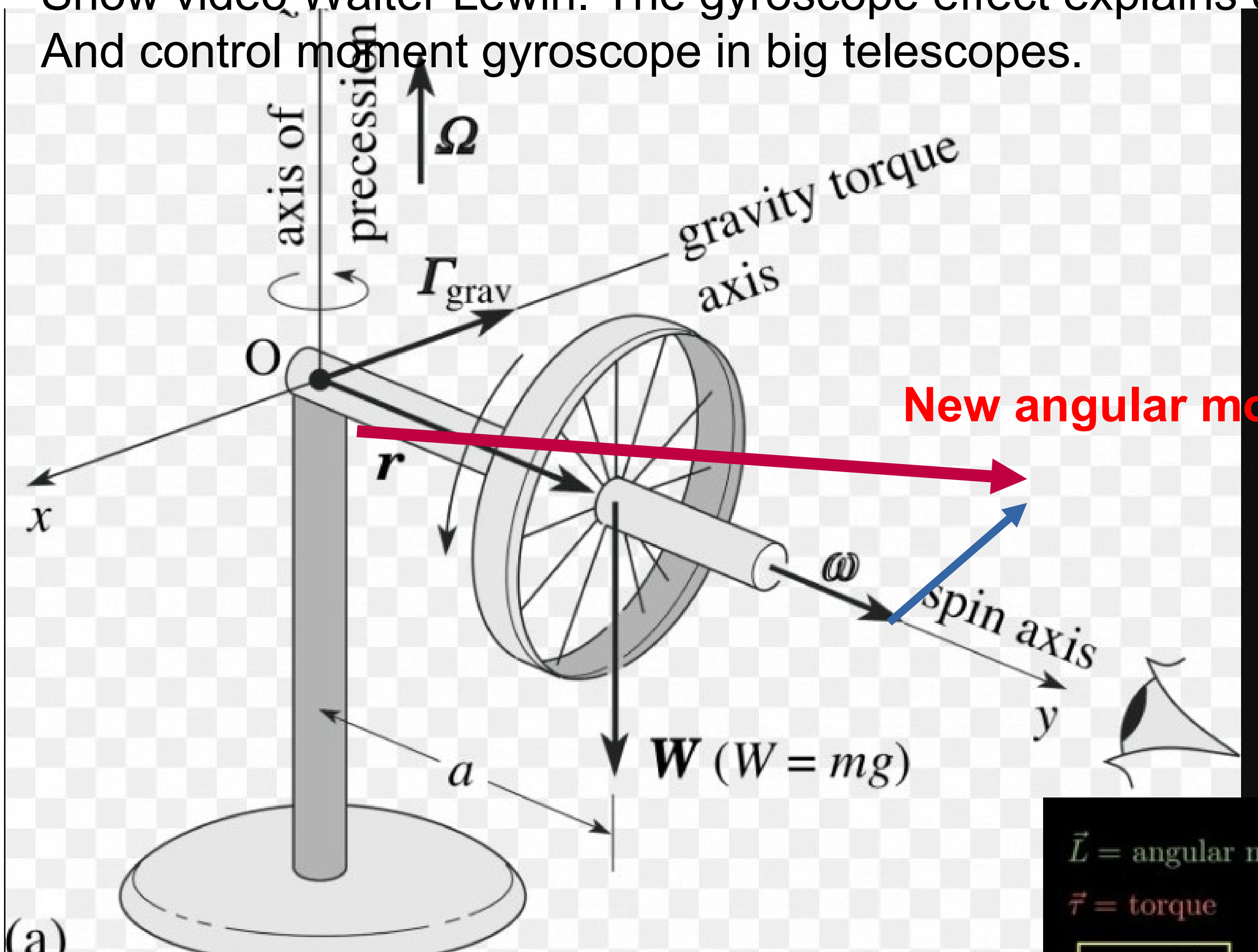

Torque changes the angular momentum

The torque that the Earth applies on the Moon
changes the angular momentum of the Moon. The Moon gains momentum
And the Earth loses momentum. The Moon moves away from us/

The Moon pulls on the bulges of the Earth (as a reaction to the pull from the Earth)
The Earth loses spin momentum and the Moon gain orbital momentum
(or you can see the bulges are ahead of Earth. There is a torque applied
To the Moon that increases its orbital momentum)
L orbital is proportional to mass and R L = m v R = m R srqrt(GM/R) = m sqrt(GMR)
So if L increases (Moon) then R increases -> Moon moves away from us

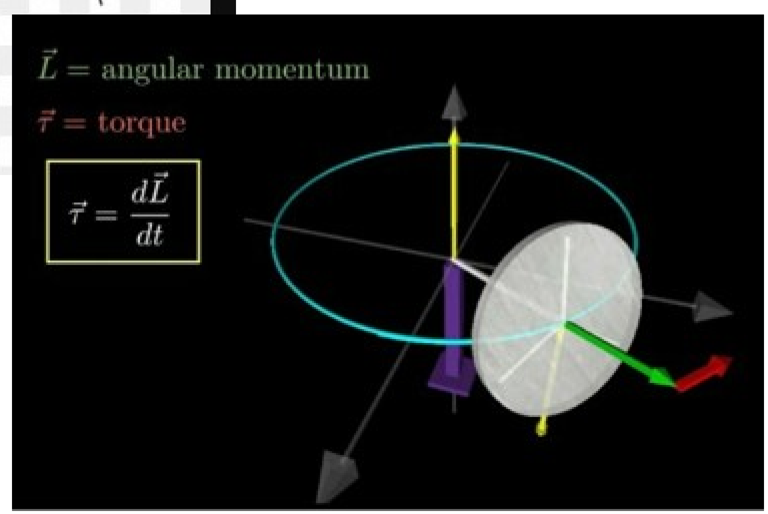
And larger R -> slower orbital speed. So Farther and slower.
The spin rotation of the Moon decreases too. As Moon and Earth are locked together.

Show video Walter Lewin. The gyroscope effect explains counter steering!
And control moment gyroscope in big telescopes.

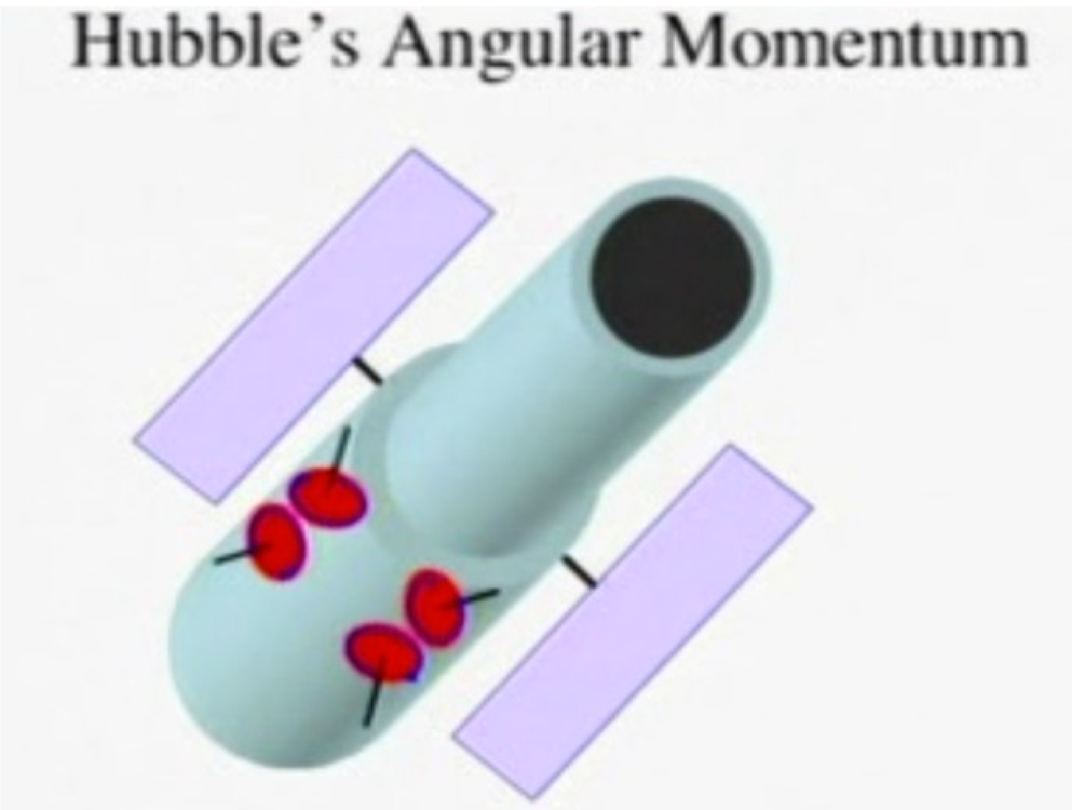


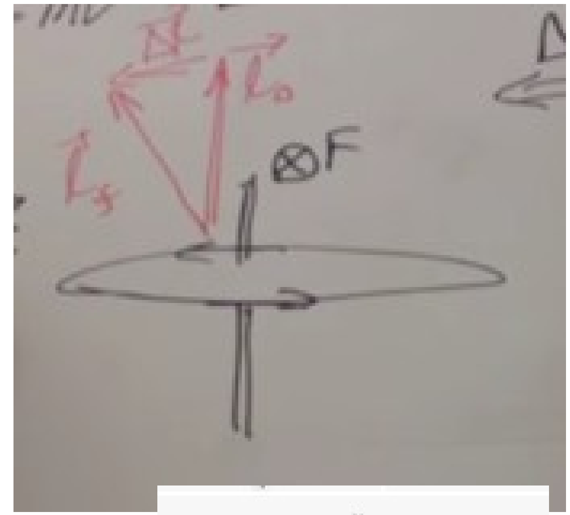

New angular momentum

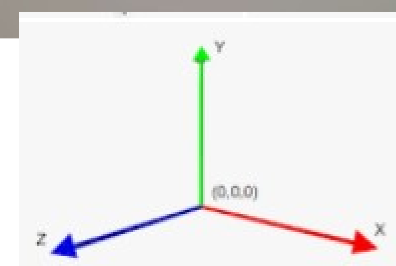
Torque produced by Weight → ΔL



The Hubble Space Telescope is equipped with **reaction wheels** to change its overall orientation, rotating the entire telescope about the speed of a minute hand on a clock – 90 degrees in 15 minutes. But to stay pointed at a single target, it uses another technology: gyroscopes. Aug 13, 2019

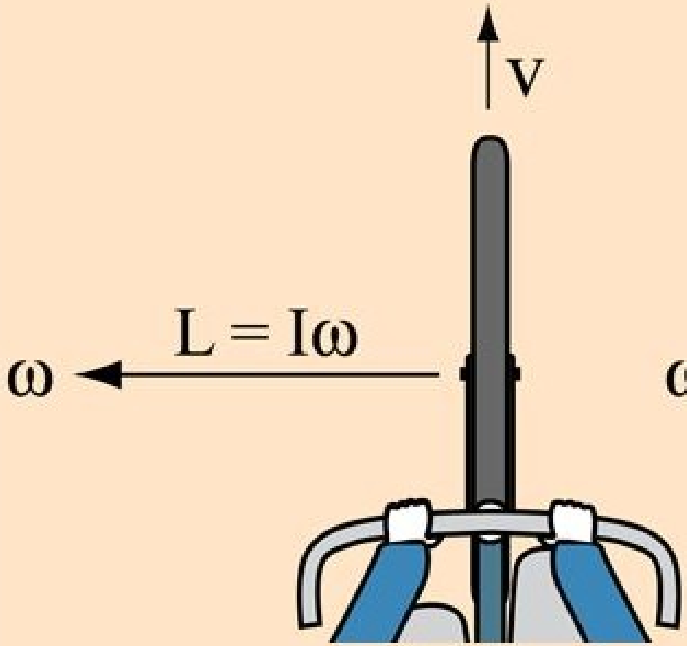


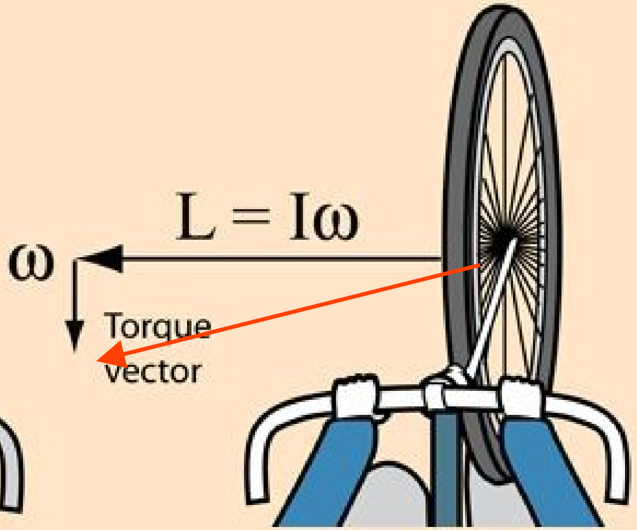



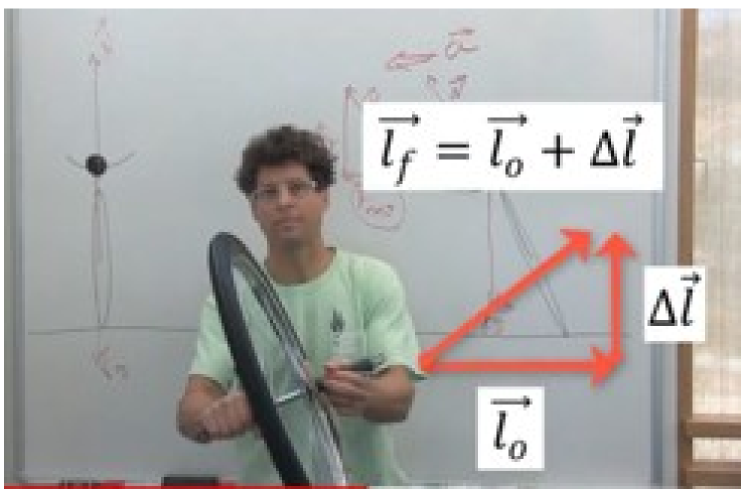

Rotation about the y-axis by force →
The wheel rotates about the zaxis

A bicycle held straight up will tend to go straight. It is tempting to say that it stabilized by the gyroscopic action of the bicycle wheels, but the gyroscopic action is quite small.

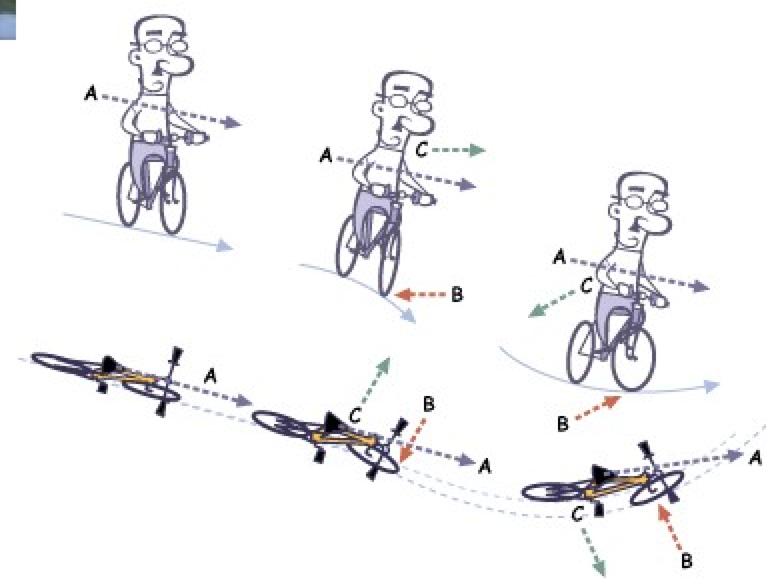



If the rider leans left, a torque will be produced which causes a counterclockwise precession of the bicycle wheel, tending to turn the bicycle to the left.











http://www.terrycolon.com/1features/counter-steering-easy.html